

Package ‘blsplotGG’

July 21, 2026

Title Plots for Seasonal Adjustment Analysts using ggplot2

Version 2.3

Description

Generates several types of time series plots useful for seasonal adjustment analysis using ggplot2. These routines rely heavily on the seasonal package to extract series and components from the 'seasonal adjustments generated by the US Census Bureau's X-13ARIMA-SEATS software, and can be generated from a single seas object or a list of seas objects. Types of plots include line plots, ratio plots, forecast plots, forecast error diagnostic plots, spectral plots, seasonal factor plots, seasonal adjustment component plots, revisions history diagnostic plots, and SEATS diagnostic plots.

Users can add grid lines, remove background shading, and shade recession regions in selected plots.

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BrailleR,
dplyr (>= 1.0.10),
forecast (>= 3.5.0),
ggplot2 (>= 3.0.0),
ggpubr,
grDevices,
lubridate,
magrittr (>= 1.5),
RColorBrewer,
Rdpack,
rlang (>= 1.1.1),
scales (>= 1.3.0),
seasonal (>= 1.10.0),
stringr (>= 1.5.0),
tidyr (>= 1.3.0),
tis,
tsbox

Suggests zoo

Depends R (≥ 3.5)

RdMacros Rdpack

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absmax	<i>Maximum absolute value of a vector</i>
--------	---

Description

Generates the maximum of the absolute value of a numeric vector.

Usage

```
absmax(x = NULL, return_abs = TRUE)
```

Arguments

- x vector of numbers. This is a required element.
- return_abs Logical scalar; if TRUE, return the absolute maximum value. If FALSE, maintain the sign of the original value. Default is TRUE.

Details

Version 2.1, 2026-05-27

Value

Maximum of the absolute value of a vector

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
r50 <- rnorm(50)
r50.absmax <- absmax(r50)
```

add_outlier_lines	<i>add lines for outliers</i>
-------------------	-------------------------------

Description

add lines for outliers to a ggplot plot object

Usage

```
add_outlier_lines(
  this_p = NULL,
  seas_obj = NULL,
  line_color = c("red", "blue", "orangered", "green", "steelblue", "blue"),
  this_palette = "Dark2",
  this_line_type = c("dashed", "dotdash", "dashed", "twodash", "dotdash", "dotdash")
)
```

Arguments

this_p	A ggplot object of a time series plot. This is a required entry.
seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddebuettel 2018). This is a required entry.
line_color	Character array of length 6; color used for different outliers, with the order being 'ao', 'ls', 'tc', 'so', 'rp', 'tls'. Default is the RColorBrewer palette "Dark2". For a listing of R color codes, see R Coder (2025).
this_palette	Character string; default RColorBrewer palette (Neuwirth 2022). Deault is "Dark2".
this_line_type	Character array of length 6; Line type used for different outliers, with the order being 'ao', 'ls', 'tc', 'so', 'rp', 'tls'. Default is c('dashed', 'dotdash', 'dashed', 'twodash', 'dotdash', 'dotdash').

Details

Version 4.5.1, 2026-06-09

Value

Revised ggplot object with lines for outliers added.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11 = "")
air_df <-
  data.frame(date = tsbox::ts_df(AirPassengers)$time,
             ori = as.double(seasonal::original(air_seas)),
             sa = as.double(seasonal::final(air_seas)),
             trend = as.double(seasonal::trend(air_seas)))
this_x11_p <- ggplot2::ggplot() +
  ggplot2::geom_line(mapping = ggplot2::aes(x=date, y = ori),
                    color = "grey", data = air_df) +
  ggplot2::geom_line(mapping = ggplot2::aes(x=date, y = sa),
                    color="steelblue", linetype="twodash", data = air_df) +
  ggplot2::geom_line(mapping = ggplot2::aes(x=date, y = trend),
                    color="darkred", linetype="twodash", data = air_df) +
  ggplot2::labs(
    title = "Airline Passenger X-11 Seasonal Adjustment",
    subtitle = NULL,
    x = "Time",
    y = "Airline Passengers")
this_x11_p_with_outlier_lines <-
  add_outlier_lines(this_x11_p, air_seas, this_palette = "Paired",
                    this_line_type = rep("dotted", 6))
```

add_recession_shade *Add shading for NBER recession dates*

Description

Add shading for US NBER recession dates to ggplot plot object. Recession dates are taken from the nberDates function of the tis package (Hallman 2021).

Usage

```
add_recession_shade(
  this_p = NULL,
  shade_color = "lightblue1",
  shade_alpha = 0.2
)
```

Arguments

<code>this_p</code>	ggplot object of a time series plot. This is a required entry.
<code>shade_color</code>	Character scalar; shading for recession region. Default is "lightblue1". For a listing of R color codes, see R Coder (2025).
<code>shade_alpha</code>	numeric scalar; controls the intensity of the shading. Default is 0.2.

Details

Version 4.2.2, 2026-06-09

Value

ggplot object with shading for recession added.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Hallman J (2021). *tis: Time Indexes and Time Indexed Series*. R package version 1.39, <https://CRAN.R-project.org/package=tis>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11 = "")
air_df <-
  data.frame(date = tsbox::ts_df(AirPassengers)$time,
             ori = as.double(seasonal::original(air_seas)),
             sa = as.double(seasonal::final(air_seas)),
             trend = as.double(seasonal::trend(air_seas)))
this_p <- ggplot2::ggplot(air_df) +
  ggplot2::geom_line(ggplot2::aes(x=date, y = ori), color = "grey") +
  ggplot2::geom_line(ggplot2::aes(x=date, y = sa), color="steelblue", linetype="twodash") +
  ggplot2::geom_line(ggplot2::aes(x=date, y = trend), color="darkred", linetype="twodash") +
  ggplot2::labs(
    title = "Airline Passenger X-11 Seasonal Adjustment",
    subtitle = NULL,
    x = "Time",
    y = "Airline Passengers")
this_p_with_recession_shading <-
  add_recession_shade(this_p, shade_color = "steelblue")
```

check_table	<i>Check table name</i>
-------------	-------------------------

Description

Check X-13ARIMA-SEATS table name so that the B1 table can be extracted when no prior adjustment is done with the seasonal package (Sax and Eddelbuettel 2018).

Usage

```
check_table(seas_obj = NULL, this_series = NULL, to_upper = FALSE)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; a table name or abbreviation from X-13ARIMA-SEATS.
to_upper	Logical scalar; return table name that is uppercase. Default is FALSE.

Details

Version 1.1, 2026-04-17

Value

Character string; an X-13ARIMA-SEATS table name. If the table is "b1" or "adjoriginal", this function will return "a1".

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas_short <-
  seasonal::seas(AirPassengers,
    series.span = ',1960.9',
    transform.function = "log",
    arima.model="(0 1 1)(0 1 1)",
    forecast.maxlead = 36,
    x11 = "",
    series.save = "b1",
    slidingspans = "")
this_series <-
  check_table(air_seas_short, "b1")
```

color_blind_palette	<i>Color-blind friendly color palette</i>
---------------------	---

Description

Color palettes that can be used that can be distinguished by color-blind people (either from Neuwirth (2022) or Chang (2025)).

Usage

```
color_blind_palette(with_grey = TRUE, brewer_palette = NULL)
```

Arguments

with_grey	Logical scalar; whether color blind palette contains 'grey', otherwise the palette contains black. Default is TRUE.
brewer_palette	Character string; a RColorBrewer palette (Neuwirth 2022). There is no default - must be a color-blind friendly palette.

Details

Version 3.2, 2026-02-10

Value

Vector of hexadecimal color codes that form a color palette that can be distinguished by color-blind people.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Chang W (2025). “Colors (ggplot2) — cookbook-r.com.” [http://www.cookbook-r.com/Graphs/Colors_\(ggplot2\)/](http://www.cookbook-r.com/Graphs/Colors_(ggplot2)/).

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

Examples

```
this_color_blind <- color_blind_palette(FALSE, brewer_palette = "Dark2")
```

convert_identify_acf	<i>Convert matrix of ACFs or PACFs generated by X-13ARIMA-SEATS identify spec to a list object</i>
----------------------	--

Description

Generates a list of the ACF or PACF generated by the identify spec

Usage

```
convert_identify_acf(seas_obj = NULL, this_plot = "iac")
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_plot	Character string; three character code for the type of plot to be generated. Allowed entries are "iac" (sample autocorrelation function, default), "ipf" (sample partial autocorrelation function).

Details

Version 3.0.1, 2026-06-09

Value

A list of matrices of ACF or PACFs produced for different orders of differencing. The list entries are named based on the orders of differencing (d0sd0 denotes no regular difference, no seasonal difference, d1sd0 denotes one regular difference, no seasonal difference, etc.)

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
shoes_identify_seas <-
  seasonal::seas(shoes2008,
    identify.diff = c(0, 1),
    identify.sdiff = c(0, 1),
    identify.save = c("iac", "ipf"),
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.maxlag = 36,
    check.acflimit = 1.96,
    check.qlimit = 0.01,
    check.print = c( 'pacf', 'pacfplot' ))
```

```
shoes_identify_acf_list <-
  convert_identify_acf(shoes_identify_seas, 'iac')
shoes_identify_pacf_list <-
  convert_identify_acf(shoes_identify_seas, 'ipc')
```

convert_spectrum_code *Convert spectrum code to visual peak code*

Description

Convert the three character code used by the plot_spectrum function so that it can be used with the visual_sig_peaks function from the blsplot package.

Usage

```
convert_spectrum_code(this_spectrum = "sp0")
```

Arguments

this_spectrum Character string; three character code for the X-13 spectrum to be generated. Allowed entries are "sp0" (modified original series), "sp1" (modified X-11 seasonally adjusted series), "sp2" (modified X-11 irregular), "s1s" (modified SEATS seasonally adjusted series), "s2s" (modified SEATS irregular), "is0" (modified composite series), "is1" (modified indirect seasonally adjusted series), "is2" (modified indirect irregular), spr (model residuals), or "ser" (extended residuals). Default: "sp0".

Details

Version 1.4, 2024-05-06

Value

Text for spectrum associated with code used in plot_spectrum function. If improper this_spectrum specified, function will return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this_code <- convert_spectrum_code("sp2")
```

difference_dates	<i>Date difference</i>
------------------	------------------------

Description

Generate the difference between two dates. Uses the lubridate package Grolemund and Wickham (2011)).

Usage

```
difference_dates(
  start_date = NULL,
  end_date = NULL,
  this_frequency = 12,
  include_start = TRUE,
  return_absolute = FALSE
)
```

Arguments

start_date	Integer array of length 2, a date where the first element is the year and the second element is the month or quarter. This is a required entry.
end_date	Integer array of length 2, a date to compare to this_date. This is a required entry.
this_frequency	Integer scalar, frequency of target time series. Entries limited to 1 (yearly), 2 (biannual), 3 (triannual), 4 (quarterly), 6 (Bimonthly), 365 (daily) and 12. Default is 12 (monthly).
include_start	Logical scalar; if TRUE, include the starting date in the difference, otherwise, the difference is left as computed. Default is TRUE.
return_absolute	Logical scalar; if TRUE, returns the absolute value of the number of days if the difference is negative

Details

Version 3.2.1, 2026-06-18

Value

Integer scalar; difference between first date and second date.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Grolemund G, Wickham H (2011). “Dates and Times Made Easy with lubridate.” *Journal of Statistical Software*, **40**(3), 1–25. <https://www.jstatsoft.org/v40/i03/>.

Examples

```
diff_start <- difference_dates(c(1990,1), c(1995,1))
```

```
display_color_blind_palettes
```

Return color-blind friendly palettes

Description

Returns the names of color palettes from the RColorBrewer package (Neuwirth 2022) that can be distinguished by color-blind people.

Usage

```
display_color_blind_palettes(this_category = NULL)
```

Arguments

`this_category` Character string; specify which category of color palette will be returned. Possible choices are "div" (diverging), "qual" (qualitative), "seq". If not specified, all color palettes are returned.

Details

Version 1.5, 2026-02-10

Value

Vector of color palette names from the RColorBrewer package that can be distinguished by color-blind people.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

Examples

```
qual_color_blind_palettes <- display_color_blind_palettes("qual")
```

employment_data_mts	<i>US Unemployment Series, four main components in an mts object</i>
---------------------	--

Description

An mts object of the four main components of US Employment expressed as time series objects that end in December, 2022.

Usage

```
data(employment_data_mts)
```

Format

An mts object with 4 time series elements in four columns:

n2000013 Employed Males 16-19

n2000014 Employed Females 16-19

n2000025 Employed Males 20+

n2000026 Employed Females 20+

employment_list	<i>US Employment Series, four main components in a list object.</i>
-----------------	---

Description

A list object of the four main components of US Employment expressed as time series objects that end in December, 2022.

Usage

```
data(employment_list)
```

Format

A list object with 4 time series elements:

n2000013 Employed Males 16-19

n2000014 Employed Females 16-19

n2000025 Employed Males 20+

n2000026 Employed Females 20+

```
extract_range_from_ggplot
```

Extract range of data from ggplot object

Description

Computes the range of all data plotted in given ggplot object.

Usage

```
extract_range_from_ggplot(this_p = NULL)
```

Arguments

`this_p` ggplot object of a time series plot. This is a required entry.

Details

Version Version 2.1, 2025-12-30

Value

Vector of length 2 with the range of the data used to generate a given ggplot object of a time series plot.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11 = "")
air_df <-
  data.frame(date = tsbox::ts_df(AirPassengers)$time,
             ori = as.double(seasonal::original(air_seas)),
             sa = as.double(seasonal::final(air_seas)),
             trend = as.double(seasonal::trend(air_seas)))
air_p <- ggplot2::ggplot(air_df, ggplot2::aes(x=date)) +
  ggplot2::geom_line(ggplot2::aes(y = ori), color = "grey") +
  ggplot2::geom_line(ggplot2::aes(y = trend), color="darkred",
                    linetype="twodash") +
  ggplot2::labs(
    title = "Airline Passenger X-11 Trend Component",
    subtitle = NULL,
    x = "Time",
    y = "Airline Passengers")
air_short_p <- plot_date_span(air_p, "1-1-1956", "1-1-1961")
air_short_range <- extract_range_from_ggplot(air_short_p)
```

flag_peak

*Flag visual significant peaks in spectra***Description**

Determine positions of visual significant peaks in spectra using the criteria of Soukup and Findley (1999).

Usage

```
flag_peak(seas_obj = NULL, spec_type = NULL, spec_freq_code = NULL)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
spec_type	Character string; type of spectrum. Possible values are 'ori', 'irr', 'rsd', 'sa'. This is a required argument.
spec_freq_code	Character string; type of frequency being tested. Possible values are 's' or 't'. This is a required argument.

Details

Version 3.2.1, 2026-06-09

Value

If visually significant peaks are found, a numeric vector of the position of the peak frequencies are returned. If no peaks found, the function returns 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Soukup RJ, Findley DF (1999). “On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment.” *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. <https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf>.

Examples

```
air_seas <- seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)', x11='')
this_flagged_peak_seas <- flag_peak(air_seas, 'ori', 's')
this_flagged_peak_td <- flag_peak(air_seas, 'ori', 't')
```

generate_alt_text	<i>Generate alt text for ggplot graphs</i>
-------------------	--

Description

Generates alt text for ggplot graph objects using the BrailleR package (Godfrey et al. 2023) or adding text as suggested by Amy Casale in her article "Writing Alt Text for Data Visualization" (Cesal 2020).

Usage

```
generate_alt_text(
  gg_object = NULL,
  chart_type = NULL,
  data_type = NULL,
  reason_text = NULL,
  short_alt = FALSE,
  BrailleR_only = FALSE
)
```

Arguments

gg_object	ggplot object from which alt text will be generated. Required entry if short_alt = FALSE.
chart_type	character scalar telling what type of plot is used in gg_object. This is a required entry.
data_type	character scalar detailing what data is used in gg_object. This is a required entry.
reason_text	character scalar detailing the reason gg_object is plotted. This is a required entry.
short_alt	logical scalar if TRUE BrailleR text will not be appended to the alt text. Default is FALSE.
BrailleR_only	logical scalar if TRUE only BrailleR text will returned. Default is FALSE.

Details

Version 4.3, 2026-02-10

Value

generate alt text for plot produced by gg_object

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Cesal A (2020). “Writing Alt Text for Data Visualization.” <https://medium.com/nightingale/writing-alt-text-for-data-visualization-2a218ef43f81>.

Godfrey AJR, Warren D, Thompson J, Murrell P, Bilton T, Sorge V (2023). *BrailleR: Improved Access for Blind Users*. R package version 1.0.2, <https://CRAN.R-project.org/package=BrailleR>.

Examples

```
air_seas <- seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11="")
air_df <-
  data.frame(date = tsbox::ts_df(AirPassengers)$time,
             ori = as.double(seasonal::original(air_seas)),
             sa = as.double(seasonal::final(air_seas)),
             trend = as.double(seasonal::trend(air_seas)))
air_p <- ggplot2::ggplot(air_df, ggplot2::aes(x=date)) +
  ggplot2::geom_line(ggplot2::aes(y = ori), color = "grey") +
  ggplot2::geom_line(ggplot2::aes(y = sa),
                    color="steelblue", linetype="twodash") +
  ggplot2::geom_line(ggplot2::aes(y = trend),
                    color="darkred", linetype="dotdash") +
  ggplot2::ggtitle("Airline Passenger X-11 Seasonal Adjustment")
air_alt_text <-
  generate_alt_text(air_p,
                  "Time series plot",
                  "International Airline Passengers time series",
                  "compare seasonal adjustment and trend to original series")
```

gen_outlier_label	<i>generate x-axis label for outliers</i>
-------------------	---

Description

Generate an x-axis label when adding lines for outliers to a ggplot plot object

Usage

```
gen_outlier_label(
  seas_obj = NULL,
  this_color = c("red", "blue", "green", "brown", "grey", "yellow")
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_color	Character array of length 6; color used for different outliers, with the order being "ao", "ls", "tc", "so", "rp", "tls". Default is c("red", "blue", "green", "brown", "grey", "yellow"). For a listing of R color codes, see R Coder (2025).

Details

Version Version 2.0.1, 2026-06-09

Value

Character string with description of outliers for x-axis label of a ggplot.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11 = "")
air_df <-
  data.frame(
    date = tsbox::ts_df(AirPassengers)$time,
    ori = as.double(seasonal::original(air_seas)),
    sa = as.double(seasonal::final(air_seas)),
    trend = as.double(seasonal::trend(air_seas)))
this_p <- ggplot2::ggplot(air_df) +
  ggplot2::geom_line(ggplot2::aes(x=date, y = ori),
    color = "grey") +
  ggplot2::geom_line(ggplot2::aes(x=date, y = sa),
    color="steelblue", linetype="twodash") +
  ggplot2::geom_line(ggplot2::aes(x=date, y = trend),
    color="darkred", linetype="twodash") +
  ggplot2::labs(
    title = "Airline Passenger X-11 Seasonal Adjustment",
    subtitle = NULL,
    y = "Airline Passengers")
this_p_with_outlier_lines <- add_outlier_lines(this_p, air_seas)
outlier_lines_label <- gen_outlier_label(air_seas)
this_p_with_outlier_lines <-
  this_p_with_outlier_lines + ggplot2::xlab(outlier_lines_label)
```

get_auto_outlier_string

Get automatic outlier names

Description

Get the names of outliers identified in the seas object for a single series.

Usage

```
get_auto_outlier_string(seas_obj = NULL, return_string = TRUE)
```

Arguments

seas_obj seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.

return_string Logical scalar; if TRUE, results returned as a character string (default). Else, results returned as a character array.

Details

Version 5.0, 2026-01-07

Value

Character string (or array if return_string is TRUE) containing a summary of the outliers identified in the regARIMA model. If no regressors or automatic outliers in the model, the routine will return a blank character (or blank vector).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <- seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11="")
this_auto_outlier <- get_auto_outlier_string(air_seas)
```

get_month_index	<i>Generate index of month abbreviation</i>
-----------------	---

Description

Process string of month abbreviation to return a numeric index .

Usage

```
get_month_index(this_month_string, full_name = FALSE)
```

Arguments

this_month_string
Character string; 3 character abbreviation of month.

full_name
Logical scalar; indicates if this is the full name of the month

Details

Version 3.1, 2026-06-15

Value

Index of month - 1 for 'Jan', 2 for 'Feb', etc. If results do not match, this function will return 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
thisOtl <- 'A02015.Jan'
thisCode <- 'A0'
thisPerChar <- substr(thisOtl,nchar(thisCode)+6,nchar(thisOtl))
thisPerIndex <- get_month_index(thisPerChar)
```

get_reg_group_string *Get types (or groups) of regressors*

Description

Generate string of names for the regressors used in the model fit for a given series.

Usage

```
get_reg_group_string(seas_obj = NULL, return_string = TRUE)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
return_string	Logical scalar; if TRUE, results returned as a character string (default). Else, results returned as a character array.

Details

Version 1.1.1, 2026-06-09

Value

Character string (or array if return_string is TRUE) containing a summary of the regressor groups in the regARIMA model. If no regressors in the model, the routine will return a blank character (or blank vector).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <- seasonal::seas(AirPassengers, slidingspans = "",
  transform.function = "log",
  x11 = "", forecast.maxlead=36,
  check.print = c( "pacf", "pacfplot" ))
air_reg_group <- get_reg_group_string(air_seas)
air_reg_group_vec <-
  get_reg_group_string(air_seas, return_string = FALSE)
```

get_reg_string	<i>Get names of regressors</i>
----------------	--------------------------------

Description

Generate string of names for the regressors used in the model fit for a given series.

Usage

```
get_reg_string(seas_obj = NULL, xreg_names = NULL, return_string = TRUE)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
xreg_names	Character vector with names of user defined regressors used in model. Default is NULL, no user defined regressors. Number of names in this vector should match number of user-defined regressors; if not, a warning message will be produced.
return_string	Logical scalar; if TRUE, results returned as a character string (default). Else, results returned as a character array.

Details

Version 4.2.1, 2026-06-09

Value

Character string (or array if return_string is TRUE) containing a summary of the regressors in the regARIMA model. If no regressors in the model, the routine will return a blank character (or blank vector).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <- seasonal::seas(AirPassengers, slidingspans = "",
                           transform.function = "log",
                           x11 = "", forecast.maxlead=36,
                           check.print = c( "pacf", "pacfplot" ))
air_reg    <- get_reg_string(air_seas)
air_reg_vec <- get_reg_string(air_seas, return_string = FALSE)
```

get_udg_index	<i>Index for entry in UDG diagnostics list</i>
---------------	--

Description

Return index for entry in UDG diagnostics list from seas object.

Usage

```
get_udg_index(udg_list = NULL, this_key = NULL)
```

Arguments

udg_list	List object generated by udg() function of the seasonal package (see Section 4 of Sax and Eddelbuettel 2018). This is a required entry.
this_key	Keyword found in udg files generated by X-13ARIMA-SEATS This is a required entry.

Details

Version 3.2, 2026-02-10

Value

An integer denoting which element in the udg output matches the key provided by the user. If there is no match, the function returns the number 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
  seasonal::seas(UKgas,
    series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    x11="",
    transform.function = "log",
    forecast.maxlead=20,
    check.print = c( "pacf", "pacfplot" ))
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_udg_index_a <- get_udg_index(ukgas_udg, this_key='a')
```

plot_acf

Generate ACF plot of the regARIMA model residuals.

Description

Generate ACF, PACF, or squared ACF plot of the regARIMA model residuals using a seas object as input. For more information on using ACF and PACF, access Box et al. (1976).

Usage

```
plot_acf(
  seas_obj = NULL,
  this_plot = "acf",
  acf_range = NULL,
  add_ci = TRUE,
  main_title = "ACF Plot",
  sub_title = NULL,
  this_x_label = "Lag",
  this_y_label = "ACF",
  this_x_axis_breaks = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  acf_color = "steelblue",
  ci_color = "grey"
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_plot	Character string; three character code for the type of plot to be generated. Allowed entries are "acf" (sample autocorrelation function, default), "pcf" (sample partial autocorrelation function), "ac2" (squared autocorrelation function).
acf_range	Range of values you wish the acf plot to be plotted over. Default is range of the series.
add_ci	Logical scalar; indicates if confidence intervals are added to the plot. Default is confidence intervals are added.
main_title	Title for the plot. Default is generated based on the value of this_plot.

sub_title	Subtitle for the plot. Default is NULL, or no subtitle.
this_x_label	Label for X axis. Default is "Lags".
this_y_label	Label for Y axis. Default is generated based on the value of this_plot.
this_x_axis_breaks	Numeric vector; sets the values for the x-axis. Default uses the value of this_frequency to set x-axis.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
acf_color	Color used for lines in ACF plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).
ci_color	Color used for confidence interval in ACF plot. Default is "grey". For a listing of R color codes, see R Coder (2025).

Details

Version 5.1.1, 2026-06-09

Value

A ggplot object that produces an ACF, PACF, or squared ACF plot of the regARIMA residuals.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Box GEP, Jenkins GM, Reinsel GC (1976). *Time Series Analysis, Forecasting and Control*, 3rd edition. Holden-Day. Series G.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 20,
    x11.seasonalma = "s3x5",
    check.print = c('pacf', 'pacfplot'),
    check.maxlag = 12,
    check.save = c("acf", "pcf", "ac2"))
p_ukgas_acf <-
  plot_acf(ukgas_x11_seas, this_plot = "acf",
    main_title = "UK Gas Model Residual ACF",
    acf_color = "darkblue")
```

```

p_ukgas_pacf <-
  plot_acf(ukgas_x11_seas, this_plot = "pcf",
    main_title = "UK Gas Model Residual PACF",
    acf_color = "darkblue")
p_ukgas_acf2 <-
  plot_acf(ukgas_x11_seas, this_plot = "ac2",
    main_title = "UK Gas Model Squared ACF",
    acf_color = "darkblue")

```

plot_acf_and_pacf	<i>Generate ACF and PACF plot of the regARIMA model residuals.</i>
-------------------	--

Description

Generate a single plot with ACF and PACF of the regARIMA model residuals using a seas object as input. For more information on using ACF and PACF, access Box et al. (1976).

Usage

```

plot_acf_and_pacf(
  seas_obj = NULL,
  add_ci = TRUE,
  overall_title = NULL,
  acf_title = "ACF Plot",
  pacf_title = "PACF Plot",
  this_x_label = "Lag",
  this_y_label = c("ACF", "PACF"),
  this_x_axis_breaks = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  acf_color = "steelblue",
  ci_color = "grey"
)

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
add_ci	Character scalar; Overall title for the plot. Default is confidence intervals are added.
overall_title	Overall title for the combined plot. Default is NULL, which doesn't produce an overall title.
acf_title	Subtitle for the ACF Plot. Default is character string 'ACF Plot'.
pacf_title	Subtitle for the PACF Plot. Default is character string 'PACF Plot'.
this_x_label	Label for X axis. Default is "Lags".
this_y_label	Character vector of length two. Labels for each Y axis. Default is c("ACF", "PACF").
this_x_axis_breaks	Numeric vector; sets the values for the x-axis. Default uses the value of this_frequency to set x-axis.

do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
acf_color	Color used for lines in ACF plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).
ci_color	Color used for confidence interval in ACF plot. Default is "grey". For a listing of R color codes, see R Coder (2025).

Details

Version 4.1.1, 2026-06-09

Value

A ggplot object that produces an ACF and PACF plot for the regARIMA model residuals.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Box GEP, Jenkins GM, Reinsel GC (1976). *Time Series Analysis, Forecasting and Control*, 3rd edition. Holden-Day. Series G.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 20,
    x11.seasonalma = "s3x5",
    check.print = c( 'pacf', 'pacfplot' ),
    check.maxlag = 12,
    check.save = c("acf", "pcf", "ac2"))
p_ukgas_acf_and_pacf <-
  plot_acf_and_pacf(ukgas_x11_seas,
    overall_title = "UK Gas Model",
    acf_color = "darkblue")
```

plot_acf_and_pacf_identify

Generate ACF and PACF plot from output of the identify spec.

Description

Generate a single plot with ACF and PACF generated by the identify spec of X-13ARIMA-SEATS for a specific set of regular and seasonal differencing. For more information about the identify spec, see Section 7.9 of U. S. Census Bureau (2025).

Usage

```
plot_acf_and_pacf_identify(
  seas_obj = NULL,
  this_diff = 0,
  this_sdiff = 0,
  add_ci = TRUE,
  overall_title = NULL,
  acf_title = "ACF Plot",
  pacf_title = "PACF Plot",
  this_x_label = "Lag",
  this_y_label = c("ACF", "PACF"),
  this_x_axis_breaks = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  acf_color = "steelblue",
  ci_color = "grey",
  remove_acf_x_labels = TRUE
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddebuettel 2018). This is a required entry.
this_diff	Integer scalar; one of the regular differences specified in the diff argument of the identify spec. Default is 0.
this_sdiff	Integer vector; one of the seasonal differences specified in the sdiff argument of the identify spec. Default is 0.
add_ci	Character scalar; Overall title for the plot. Default is confidence intervals are added.
overall_title	Overall title for the combined plot. Default is a text string showing the orders of differencing selected.
acf_title	Subtitle for the ACF Plot. Default is character string 'ACF Plot'.
pacf_title	Subtitle for the PACF Plot. Default is character string 'PACF Plot'.
this_x_label	Label for X axis. Default is "Lags".
this_y_label	Character vector of length two. Labels for each Y axis. Default is c("ACF", "PACF").

this_x_axis_breaks	Numeric vector; sets the values for the x-axis. Default uses the value of this_frequency to set x-axis.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
acf_color	Color used for lines in ACF plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).
ci_color	Color used for confidence interval in ACF plot. Default is "grey". For a listing of R color codes, see R Coder (2025).
remove_acf_x_labels	Logical scalar; indicates if X-axis tick labels are removed (TRUE) or printed (FALSE). Default is TRUE.

Details

Version 6.1.1, 2026-06-09

Value

A ggplot object that produces an ACF and PACF plot for the original series with user-specified orders of differencing.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_identify_seas <-
  seasonal::seas(shoes2008,
    identify.diff = c(0, 1),
    identify.sdiff = c(0, 1),
    identify.save = c("iac", "ipc"),
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.maxlag = 36,
    check.acflimit = 1.96,
    check.qlimit = 0.01,
```

```

        check.print = c( 'pacf', 'pacfplot' ))
p_shoes_acf_and_pacf_identify_d0_sd0  <-
  plot_acf_and_pacf_identify(shoes_identify_seas,
    overall_title = "US Shoe Sales - No Differencing",
    acf_color = "darkblue")
p_shoes_acf_and_pacf_identify_d1_sd1  <-
  plot_acf_and_pacf_identify(shoes_identify_seas,
    this_diff = 1,
    this_sdiff = 1,
    overall_title = "US Shoe Sales - Regular and Seasonal Differencing",
    acf_color = "darkblue",
    remove_acf_x_labels = FALSE)

```

plot_acf_matrix	<i>Generate ACF plot from a matrix of the ACF.</i>
-----------------	--

Description

Generate ACF, PACF, or squared ACF plot of the regARIMA model residuals (or the transformed original series in the case of the identify output of X-13ARIMA-SEATS) from a matrix of the ACF. This matrix can be extracted from a seas object produced by the seasonal package (Sax and Eddelbuettel 2018).

Usage

```

plot_acf_matrix(
  acf_matrix = NULL,
  acf_range = NULL,
  add_ci = TRUE,
  acflimit = 1.6,
  main_title = "ACF Plot",
  sub_title = NULL,
  this_x_label = "Lag",
  this_y_label = "ACF",
  this_x_axis_breaks = NULL,
  this_frequency = 12,
  acf_color = "steelblue",
  ci_color = "grey"
)

```

Arguments

acf_matrix	Numeric matrix containing the ACF, PACF, or squared ACF with columns with SE, Ljung Box Q, lags, if associated with the file. This is a required entry.
acf_range	Range of values you wish the acf plot to be plotted over. Default is range of the series.
add_ci	Logical scalar; indicates if confidence intervals are added to the plot. Default is confidence intervals are added.
acflimit	Numeric scalar; the multiplier for the confidence interval usually read from the udg. Default: 1.6.
main_title	Title for the plot. Default is "ACF Plot".

sub_title	Subtitle for the plot. Default is NULL, or no subtitle.
this_x_label	Label for X axis. Default is "Lags".
this_y_label	Label for Y axis. Default is "ACF".
this_x_axis_breaks	Numeric vector; sets the values for the x-axis. Default uses the value of this_frequency to set x-axis.
this_frequency	Integer scalar; Frequency of the time series. Default is 12.
acf_color	Color used for lines in ACF plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).
ci_color	Color used for confidence interval in ACF plot. Default is "grey". For a listing of R color codes, see R Coder (2025).

Details

Version 4.3.1, 2026-06-09

Value

A ggplot object that produces an ACF, PACF, or squared ACF plot of the regARIMA residuals.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 20,
    x11.seasonalma = "s3x5",
    check.print = c( 'pacf', 'pacfplot' ),
    check.maxlag = 12,
    check.save = c("acf", "pcf", "ac2"))
ukgas_acf_matrix <-
  seasonal::series(ukgas_x11_seas, "acf")
p_ukgas_acf_matrix <-
  plot_acf_matrix(ukgas_acf_matrix,
    main_title = "UK Gas Residual ACF",
    this_frequency = 4,
    acf_color = "darkblue")
```

plot_all_trend_lags *History Plot of All Trend Revisions*

Description

Generates a ggplot2 object of estimates from a revisions history of a trend component for a given series, incorporating all trend lag revisions. For more information on revisions history diagnostics, see Sections 6.3 and 7.8 of U. S. Census Bureau (2025).

Usage

```
plot_all_trend_lags(
  seas_obj = NULL,
  this_trend_name = NULL,
  main_title = "All Trend Revision Plot",
  sub_title = NULL,
  this_y_label = NULL,
  this_x_label = "Time",
  do_grid = FALSE,
  do_background = FALSE,
  base_color = "darkblue",
  whisker_color = "darkgrey"
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_trend_name	Character string; name of time series. No default.
main_title	Title for the plot. Default is character string 'Trend History Plot'.
sub_title	Subtitle for the plot. Default is NULL.
this_y_label	Label for Y axis. Default is "Ratio".
this_x_label	Label for X axis. Default is "Time".
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
base_color	Character scalar for plot of the initial trend. Default is "darkblue". For a listing of R color codes, see R Coder (2025).
whisker_color	Character scalar for color used for lines representing lagged trend estimates in the trend history plot. Default is "darkgrey"

Details

Version 3.2.1, 2026-06-09

Value

A ggplot object that produces a history plot of the trend component. The seas object in the seas_obj argument must contain output for a revisions history analysis for trends with trendlags set to some values - these lags should be in sequence, such as history.trendlags = 1:4.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_seas <-
  seasonal::seas(shoes2008, x11.save = "d13",
    x11 = "", transform.function = "log",
    check.print = c("none", "+acf", "+acfplot", "+normalitytest"),
    regression.aictest = c("td", "easter"),
    regression.save = c("td", "hol"),
    outlier.types = "all",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 60,
    history.estimates = c("sadj", "sadjchng", "trend"),
    history.trendlags = 1:4,
    history.print = "all",
    history.save = c("tre", "trr"))
p_shoes_trend_history <-
  plot_all_trend_lags(shoes_seas, this_trend_name = "shoes",
    main_title = "All Trend Revisions, Lags 1 to 4",
    sub_title = "US Retail Sales of Shoes")
```

plot_changes_history *Revisions History Plot for Seasonal Adjustment Changes*

Description

Generates a ggplot object of estimates from a revisions history of either the seasonal adjustment or trend changes for a given series from a seas object. For more information on revisions history diagnostics, see Sections 6.3 and 7.8 of U. S. Census Bureau (2025).

Usage

```
plot_changes_history(
  seas_obj = NULL,
  this_change = "sa",
  plot_start = NULL,
  main_title = "SA Change History Graph",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  line_color = NULL,
  this_palette = "Paired",
  this_legend_title = "Filter",
  this_legend_position = "right",
  legend_title_size = 12,
  legend_text_size = 10,
  this_legend_space = NULL
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_change	Character string that determines whether the revisions of the changes of the seasonally adjustment ("sa") or the trend ("trend") is plotted. The default is "sadj".
plot_start	Integer vector of length two. Start of the plot. Default is NULL, which defaults to the start of the history analysis.
main_title	Title for the plot. Default is character string 'SA Change History Graph'.
sub_title	Subtitle for the plot. Default is NULL.
this_x_label	Label for X axis. Default is "Time".
this_y_label	Label for Y axis. Default is NULL.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Vector with colors used for lines in history plot. Should be of length 2. For a listing of R color codes, see R Coder (2025). Default is NULL, which indicates that the palette specified in this_palette is used to generate colors for this plot.
this_palette	Colors used for lines in this plot. Default is the RColorBrewer palette "Paired" (Neuwirth 2022).
this_legend_title	Title for legend. Default is "Series"
this_legend_position	Character string; indicates the position of the plot legend. Possible values are "right" (default), "left", "top", "bottom", and "none"

legend_title_size
integer scalar; Size of the legend title. Default is 12.

legend_text_size
integer scalar; Size of the legend title. Default is 10.

this_legend_space
integer scalar; Space between legend and plot in points. Default is determined by ggplot2.

Details

Version 4.3.1, 2026-06-09

Value

A ggplot object that produces a history plot of the seasonal adjustment or trend changes. The seas object in the seas_obj argument must contain output for a revisions history analysis for seasonal adjustment changes.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_seas <-
  seasonal::seas(shoes2008, x11 = "",
    transform.function = "log",
    regression.aictest = c("td", "easter"),
    outlier.types = "all",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 60,
    history.estimates =
  c("sadj", "sadjchn", "trend", "trendchn"),
    history.sadjlags = c(1,12),
    history.trendlags = c(1,2,3),
    history.print = "all",
    history.save = c("che", "tce"))
p_shoes_sadjchn_history <-
  plot_changes_history(shoes_seas, plot_start = c(2003, 1),
    main_title = "Seasonal Adjustment Change History Graph",
```

```

      sub_title = "US Retail Sales of Shoes")
p_shoes_trendchng_history <-
  plot_changes_history(shoes_seas,
    this_change = "trend",
    plot_start = c(2003, 1),
    main_title = "Trend Change History Graph",
    sub_title = "US Retail Sales of Shoes")

```

plot_component_facet *Component facet plot*

Description

Generates a plot for the components of a seasonal adjustment - trend, seasonal, and irregular - from a seas object generated from a call of seas function of the seasonal package (Sax 2018).

Usage

```

plot_component_facet(
  seas_obj = NULL,
  main_title = "Component Facet Plot",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = " ",
  do_grid = FALSE,
  do_background = FALSE,
  this_scale = "free_y",
  line_color = "steelblue"
)

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Edelbuettel 2018). This is a required entry.
main_title	Title for the plot. Default is character string 'Component Facet Plot'.
sub_title	Subtitle for the plot. Default is either "SEATS seasonal adjustment" or "X-11 seasonal adjustment".
this_x_label	Label for X axis. Default is "Time".
this_y_label	Label for Y axis. Default is " ".
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
this_scale	Character scalar; set the scale for the y-axis of the facet plots. Default is "free_y". To set axis to be the same for the different plots, set this_scale = "fixed". Note: Setting this_scale = "fixed" seasonal is not recommended for multiplicative adjustments.
line_color	Color used for lines in component plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).

Details

Version 1.5.1, 2026-06-09

Value

A ggplot object that generates a facet plot for the trend, seasonal, and irregular from the adjustment encapsulated in seas_obj.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS. R package version 1.7.* <https://CRAN.R-project.org/package=seasonal>.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
shoes_x11_seas <-
  seasonal::seas(shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 36,
    regression.aictest = c("td", "easter"),
    regression.save = c("td", "hol"),
    x11.seasonalma = "s3x5", check.print = c( 'pacf', 'pacfplot' ))
shoes_comp_plot <-
  plot_component_facet(shoes_x11_seas,
    main_title = "US Retail Shoe Sales",
    sub_title = "X-11 (3x5 Seasonal filter)")
```

plot_date_span

Plot a span of data

Description

Shortens the time span of an existing time series ggplot object by limiting the X axis to user specified dates. Uses the lubridate package Golemund and Wickham (2011)).

Usage

```
plot_date_span(
  this_p = NULL,
  this_start_span = NULL,
  this_end_span = NULL,
  this_date_breaks = "1 year",
```

```

    this_date_format = "%Y",
    reset_y_limit = FALSE
  )

```

Arguments

this_p A ggplot object of a time series plot. This is a required entry.

this_start_span Character scalar with the date of the start of the span to be plotted. This is a required entry.

this_end_span Character scalar with the date of the end of the span to be plotted. This is a required entry.

this_date_breaks Character scalar with the interval for tic marks on the x-axis. Default is '1 year'.

this_date_format Character scalar with the format used for the date label on the x-axis. Default is '%Y'.

reset_y_limit Logical scalar that if TRUE, the range of the y-axis is reset. Default is FALSE.

Details

Version 2.4.1, 2026-06-18

Value

A ggplot object that produces a subplot of the submitted plot.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Grolemund G, Wickham H (2011). "Dates and Times Made Easy with lubridate." *Journal of Statistical Software*, **40**(3), 1–25. <https://www.jstatsoft.org/v40/i03/>.

Examples

```

air_seas <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11 = "")
air_df <-
  data.frame(date = tsbox::ts_df(AirPassengers)$time,
             ori = as.double(seasonal::original(air_seas)),
             sa = as.double(seasonal::final(air_seas)),
             trend = as.double(seasonal::trend(air_seas)))
air_p <- ggplot2::ggplot(air_df, ggplot2::aes(x=date)) +
  ggplot2::geom_line(ggplot2::aes(y = ori), color = "grey") +
  ggplot2::geom_line(ggplot2::aes(y = trend),
                    color="darkred",
                    linetype="twodash") +
  ggplot2::labs(
    title = "Airline Passenger X-11 Trend Component",
    subtitle = NULL,

```

```

x = "Time",
y = "Airline Passengers")
air_short_p <-
  plot_date_span(air_p, "1-1-1956", "12-31-1960",
    reset_y_limit = TRUE)

```

plot_double_spectrum	<i>Generate double spectrum plot of the original and seasonally adjusted series.</i>
----------------------	--

Description

Generate plot of spectrum of original series and seasonally adjusted series on same axis. For more information on spectral diagnostics, see Sections 6.1 and 7.17 of U. S. Census Bureau (2025).

Usage

```

plot_double_spectrum(
  seas_obj = NULL,
  xaxis_bls = TRUE,
  main_title = NULL,
  series_name = NULL,
  line_color = c("darkblue", "darkgreen"),
  median_color = c("blue", "green"),
  freq_color = c("steelblue", "forestgreen"),
  peak_color = c("violet", "brown")
)

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
xaxis_bls	Logical scalar; indicates if x-axis of spectral plot will be frequency by month rather than the actual frequencies. Default sets x-axis to frequency by month.
main_title	Character string; main title of plot. Default is 'AR Spectrum'; if series_name is specified, the series name is added to the end of main_title if that argument is not specified.
series_name	Character scalar; name of the time series used in seas_obj. Used as the label of the Y-axis if specified.
line_color	Character vector of length 2. Colors used for original and seasonally adjusted spectrum in plot. Defaults are c("darkblue", "darkgreen"). For a listing of R color codes, see R Coder (2025).
median_color	Character vector of length 2. Colors used for medians of the original and seasonally adjusted spectrum, respectively. Defaults are c("blue", "green").
freq_color	Character vector of length 2. Colors used for seasonal and trading day frequencies, respectively. Defaults are c("steelblue", "forestgreen").
peak_color	Character vector of length 2. Colors used for peaks at seasonal and trading day frequencies, respectively. Defaults are c("violet", "brown").

Details

Version 4.3, 2026-07-06

Value

ggplot object of spectrum of original series and seasonally adjusted series on same axis.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, x11="",
    arima.model = "(0 1 1)(0 1 1)",
    spectrum.save = c("sp0", "sp1"))
plot_double_spectrum_air <-
  plot_double_spectrum(air_seas,
    series_name = "AirPassengers",
    line_color = c("steelblue", "forestgreen"),
    median_color = c("blue", "green"),
    freq_color = c("darkblue", "darkgreen"),
    peak_color = c("red", "orange"))
```

plot_fcst

Forecast plot

Description

Generates a ggplot plot of regARIMA forecasts with confidence bounds.

Usage

```
plot_fcst(
  seas_obj = NULL,
  main_title = "ARIMA forecasts",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = " ",
```

```

length_ori = 2,
do_grid = FALSE,
do_background = FALSE,
this_palette = c("steelblue", "blue", "darkgrey", "darkgrey"),
this_legend_title = "Forecast",
this_legend_title_size = 12,
this_legend_text_size = 10,
this_legend_position = "right",
this_legend_space = 10
)

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
main_title	Character string; main title of plot. Default is 'ARIMA forecasts'.
sub_title	Subtitle for the plot. Default is to generate the subtitle with the value of the average absolute percentage error in within-sample forecasts (AAPE) for the last three years.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "
length_ori	Integer scalar; number of years of the original series to show with forecasts. Default is 2 years.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
this_palette	Array of character strings; color used for original series, forecast, and upper and lower forecast bounds. Default is c("steelblue", "blue", "darkgrey", "darkgrey"). For a listing of R color codes, see R Coder (2025).
this_legend_title	Title for legend. Default is "Forecast"
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.
this_legend_text_size	integer scalar; Size of the legend title. Default is 10.
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 4.3.1, 2026-06-09

Value

A ggplot object of the regARIMA forecasts with confidence bounds.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <- seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)",
                           forecast.maxlead = 60, forecast.save = "fct",
                           series.save = "a1")
air_fcst_p <-
  plot_fcst(air_seas,
            main_title = "Forecasts for Airline Passengers", do_grid = TRUE)
```

plot_fcst_history	<i>Generate forecast history plot</i>
-------------------	---------------------------------------

Description

Generate forecast history plot, which compares the sum of squared forecast errors for two models. This follows the theory established in Findley (2005). For more information on history analysis, see Chapters 6.3 and 7.8 of U. S. Census Bureau (2025).

Usage

```
plot_fcst_history(
  seas_md1 = NULL,
  seas_md2 = NULL,
  main_title = "Sum of Squared Forecast Error Differences",
  name_md1 = "Model 1",
  name_md2 = "Model 2",
  do_grid = FALSE,
  do_background = FALSE,
  this_x_label = "Time",
  this_y_label = " ",
  this_palette = c("steelblue", "darkgreen"),
  this_legend_title = "Fcst History",
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

seas_md11	seas object generated from a call of seas on a single time series for the first model This is a required entry.
seas_md12	seas object generated from a call of seas on a single time series for the second model This is a required entry.
main_title	Character string; main title of plot. Default is 'Sum of Squared Forecast Error Differences'.
name_md11	Character string; Description of first model for use in the subtitle. Default is 'Model 1'.
name_md12	Character string; Description of second model for use in the subtitle. Default is 'Model 2'.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "
this_palette	Character array of length 2; color used for each forecast lag. Default is c("steelblue", "darkgreen").
this_legend_title	Title for legend. Default is "Fcst History"
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.
this_legend_text_size	integer scalar; Size of the legend title. Default is 10.
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 4.3, 2026-05-09

Value

A ggplot object that produces a forecast history plot. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Findley DF (2005). “Asymptotic stationarity properties of out-of-sample forecast errors of mis-specified regARIMA models and the optimality of GLS for one-step-ahead forecasting.” *Statistica Sinica*, **15**(2), 447–476.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
air_seas_md1 <-
  seasonal::seas(AirPassengers, x11="", slidingspans = "",
    transform.function = "log", regression.aictest = NULL,
    arima.model = "(0 1 1)(0 1 1)", outlier = NULL,
    forecast.maxlead = 36, check.print = c( 'pacf', 'pacfplot' ),
    history.fstep = c(1, 12), history.estimates = 'fcst',
    history.save = 'fcsterrors')
air_seas_md12 <-
  seasonal::seas(AirPassengers, x11="", slidingspans = "",
    transform.function = "log",
    arima.model = "(0 1 1)(0 1 1)", regression.variables = c("td"),
    forecast.maxlead = 36, check.print = c( 'pacf', 'pacfplot' ),
    history.fstep = c(1, 12), history.estimates = 'fcst',
    history.save = 'fcsterrors')
plot_fcst_history_air <-
  plot_fcst_history(air_seas_md1, air_seas_md12,
    main_title = 'Sum of Squared Forecast Error Differences for Airline Passengers',
    name_md11 = 'Airline model', name_md12 = 'Airline model + regressors')
```

plot_fcst_two

Forecast plot for two models

Description

Generates plot that compares regARIMA forecasts for two models of the same series. The forecasts are extracted from two seas objects from the same series generated from the seas function of the seasonal package (Sax 2018).

Usage

```
plot_fcst_two(
  seas_obj_one = NULL,
  seas_obj_two = NULL,
  main_title = "ARIMA forecasts",
  sub_title = NULL,
  name_md11 = "Model 1",
  name_md12 = "Model 2",
  this_x_label = "Time",
  this_y_label = " ",
  length_ori = 2,
  do_grid = FALSE,
```

```

do_background = FALSE,
this_palette = c("darkgrey", "steelblue", "darkgreen"),
this_legend_title = "Forecast",
this_legend_title_size = 12,
this_legend_text_size = 10,
this_legend_position = "right",
this_legend_space = 10
)

```

Arguments

seas_obj_one	seas object generated from a call of seas on a single time series This is a required entry.
seas_obj_two	seas object generated from a call of seas on the same time series, but a different regARIMA model. This is a required entry.
main_title	Character string; main title of plot. Default is 'ARIMA Residuals'.
sub_title	Subtitle for the plot. Default is to generate the subtitle.
name_md11	Character string; Description of first model for use in the subtitle. Default is 'Model 1'.
name_md12	Character string; Description of second model for use in the subtitle. Default is 'Model 2'.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "
length_ori	Integer scalar; number of years of the original series to show with forecasts. Default is 2 years.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
this_palette	Array of character strings; color used for original series, forecast, and upper and lower forecast bounds. Default is c("darkgrey", "steelblue", "darkgreen"). For a listing of R color codes, see R Coder (2025).
this_legend_title	Title for legend. Default is "Forecast"
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.
this_legend_text_size	integer scalar; Size of the legend title. Default is 10.
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 3.1.1, 2026-06-09

Value

A ggplot object of the regARIMA forecasts for two models of the same series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS. R package version 1.7.* <https://CRAN.R-project.org/package=seasonal>.

Examples

```
shoes_seas_one <-
  seasonal::seas(shoes2008, x11="", slidingspans = "", transform.function = "log",
    arima.model = "(0 1 1)(0 1 1)", regression.aictest = NULL,
    outlier = NULL, forecast.maxlead = 60, forecast.save = "fct",
    check.print = c( 'pacf', 'pacfplot' ), series.save = "a1")
shoes_seas_two <-
  seasonal::seas(shoes2008, x11="", slidingspans = "", transform.function = "log",
    arima.model = "(0 1 1)(0 1 1)", regression.variables = c("td"),
    forecast.maxlead = 60, forecast.save = "fct",
    check.print = c( 'pacf', 'pacfplot' ), series.save = "a1")
shoes_fcst_two_p <-
  plot_fcst_two(shoes_seas_one, shoes_seas_two,
    main_title = "Forecast Comparison Plot",
    name_md11 = 'Airline', name_md12 = 'Airline + reg',
    do_grid = TRUE)
```

plot_first_difference *First Difference Plot*

Description

Generates a ggplot2 object of the first difference of a time series grouped by months or quarters. The series must be a ts object.

Usage

```
plot_first_difference(
  this_series = NULL,
  take_log = FALSE,
  main_title = "First Difference Plot",
  remove_y_axis = TRUE,
  x_title_size = 10,
  this_text_size = 2.5,
  this_text_color = "steelblue",
  add_zero_line = FALSE
)
```

Arguments

<code>this_series</code>	ts object of a single time series This is a required entry.
<code>take_log</code>	Logical scalar, specifies that a log transformation will be taken before differencing. Default is FALSE
<code>main_title</code>	Title for the plot. Default is character string 'First Difference Plot'.
<code>remove_y_axis</code>	Logical scalar. If TRUE, removes the y-axis labels and tick marks from all subplots. Default is FALSE, which keeps the y-axis labels and tick marks.
<code>x_title_size</code>	Integer scalar, size of the x-axis title. Default is 10.
<code>this_text_size</code>	Integer scalar, size of the plotting characters. Default is 2.5.
<code>this_text_color</code>	Character scalar, color of the plotting characters. Default is "steelblue". For a listing of R color codes, see R Coder (2025).
<code>add_zero_line</code>	Logical scalar. If TRUE, adds a horizontal line at zero for all plots. Default is FALSE, no horizontal reference line.

Details

Version 2.3.1, 2026-06-09

Value

A ggplot object that produces a plot of the first differences of a series specified by `this_series` grouped by month or quarter. The time series specified should be either a monthly or quarterly series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
shoes_seas <-
  seasonal::seas(shoes2008, x11.save = "d13",
    x11 = "", transform.function = "log",
    check.print = c("none", "+acf", "+acfplot", "+normalitytest"),
    regression.aictest = c("td", "easter"),
    regression.save = c("td", "hol"),
    outlier.types = "all",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 60)
shoesSA <- seasonal::final(shoes_seas)
p_shoes_sa_diff <-
  plot_first_difference(shoesSA,
    main_title = "Seasonal Adjustment First Differences",
    remove_y_axis = FALSE, x_title_size = 10,
    add_zero_line = TRUE)
```

plot_fts

*Final t-statistics for the outlier identification procedure plot***Description**

Generates a plot of the final t-statistics for the outlier identification procedure. A description of the outlier identification procedure is given in the documentation of the outlier spec in Section 7.11 of U. S. Census Bureau (2025), with more details in Appendix B of Findley et al. (1998).

Usage

```
plot_fts(
  seas_obj = NULL,
  start_plot = NULL,
  main_title = "Outlier T-Values",
  this_y_label = NULL,
  this_x_label = "Time",
  add_identified_outliers = FALSE,
  do_grid = FALSE,
  do_background = FALSE,
  plot_abs = TRUE,
  color_otl = NULL,
  this_palette = "Dark2",
  this_legend_title = "Outlier",
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
start_plot	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
main_title	Character string; main title of plot. Default is 'Outlier T-Values'.
this_y_label	Character string; y-axis label for plot, if specified.
this_x_label	Label for X axis. Default is "Time".
add_identified_outliers	Logical scalar; indicates if outlier plots will include identified outliers. Default is not including identified outliers.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
plot_abs	Logical scalar; if TRUE, plot the maximum absolute value of the t-stats. If FALSE, maintain the sign of the original t-statistic in the plot. Default is TRUE.

color_otl	Character array of length 3; color used for different outliers, with the order being 'ao', 'ls', 'tc'. Default is NULL. For a listing of R color codes, see R Coder (2025).
this_palette	Character string; default RColorBrewer palette (Neuwirth 2022). Deault is "Dark2".
this_legend_title	Title for legend. Default is "Year".
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.
this_legend_text_size	integer scalar; Size of the legend title. Default is 10.
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 7.2.1, 2026-06-09

Value

A ggplot object which produces a plot of the final t-statistics from the automatic outlier identification procedure.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

- Findley DF, Monsell BC, Bell WR, Otto MC, Chen BC (1998). "New capabilities of the X-12-ARIMA seasonal adjustment program (with discussion)." *Journal of Business and Economic Statistics*, **16**, 127–77. <https://www.census.gov/content/dam/Census/library/working-papers/1998/adrm/jbes98.pdf>.
- Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.
- R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].
- Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.
- U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
air_seas_outlier <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)",
    outlier.save = "fts", outlier.types = "all")
plot_fts_air <-
  plot_fts(air_seas_outlier,
    main_title = "Outlier T-Values for Airline Passengers")
plot_fts_air_add_id_otl <-
  plot_fts(air_seas_outlier,
    main_title = "Outlier T-Values for Airline Passengers",
    add_identified_outliers = TRUE)
plot_fts_air_no_abs_add_id_otl <-
  plot_fts(air_seas_outlier, plot_abs = FALSE,
    main_title = "Outlier T-Values for Airline Passengers",
    add_identified_outliers = TRUE)
```

```
plot_fully_differenced_transformed
```

*Plot fully differenced transformed series from an X-13ARIMA-SEATS
SEATS seasonal adjustment run.*

Description

Generate plot of the fully differenced transformed series from a SEATS adjustment from a seas object generated by the seasonal package.

Usage

```
plot_fully_differenced_transformed(
  seas_obj = NULL,
  this_series = "dor",
  main_title = NULL,
  sub_title = NULL,
  this_y_label = NULL,
  y_limit = NULL,
  this_x_label = "Time",
  start_plot = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  draw_recess = FALSE,
  recess_color = "lightgrey",
  recess_sub = TRUE,
  title_size = 10,
  add_zero_line = FALSE,
  line_color = "steelblue"
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
----------	--

<code>this_series</code>	Character string; series for which SEATS produces a fully differenced transformed series, limited to the original series (<code>dor</code> , the default), the seasonally adjusted series (<code>dsa</code>), or the trend component (<code>dtr</code>). For other entries, the function will print an error message and return a <code>NULL</code> . For more information on SEATS input options in X-13ARIMA-SEATS, (see Section 7.14 of U. S. Census Bureau 2025).
<code>main_title</code>	Character string; main title of plot. A title will be generated if no title is specified.
<code>sub_title</code>	Character string; subtitle of plot. There is no default subtitle.
<code>this_y_label</code>	Character string; y-axis label for plot, if specified.
<code>y_limit</code>	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the series specified.
<code>this_x_label</code>	Label for X axis. Default is "Time".
<code>start_plot</code>	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
<code>do_grid</code>	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
<code>do_background</code>	Logical scalar; indicates grey background included in plot. Default is no grey background;
<code>draw_recess</code>	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates. Default is no recession shading.
<code>recess_color</code>	Character string; color used for shading of recession region. Default is 'lightgrey'. For a listing of R color codes, see R Coder (2025).
<code>recess_sub</code>	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label.
<code>title_size</code>	integer scalar; Size of the plot title. If <code>this_series = "all"</code> , the overall title size will be <code>title_size + 2</code> . Default is 10.
<code>add_zero_line</code>	Logical scalar; if <code>TRUE</code> , adds a grey zero line to plots. Default is <code>FALSE</code> , no zero line is added.
<code>line_color</code>	Character string; color used for series in the plot. Default is 'steelblue'. For a listing of R color codes, see R Coder (2025).

Details

Version 4.3.1, 2026-06-09

Value

A `ggplot` object that generates a plot of the fully differenced transformed series from a SEATS seasonal adjustment. If SEATS seasonal adjustment not produced, print out error message and return `NULL`.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

- R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].
- Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.
- U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_seats_seas <-
  seasonal::seas(shoes2008, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    seats.finite = "yes",
    seats.save = c( 'dor', 'dsa', 'dtr' ) )
p_ori_full_diff <-
  plot_fully_differenced_transformed(shoes_seats_seas, "dor",
    sub_title = "US Shoe Sales",
    do_grid = TRUE,
    line_color = "darkgreen")
p_sa_full_diff <-
  plot_fully_differenced_transformed(shoes_seats_seas, "dsa",
    sub_title = "US Shoe Sales",
    do_grid = TRUE,
    line_color = "darkgreen")
p_trn_full_diff <-
  plot_fully_differenced_transformed(shoes_seats_seas, "dtr",
    sub_title = "US Shoe Sales",
    do_grid = TRUE,
    line_color = "darkgreen")
```

```
plot_fully_differenced_transformed_all
```

Plot fully differenced transformed series for the original series, seasonal adjustment, and trend from an X-13ARIMA-SEATS SEATS seasonal adjustment run.

Description

Generate plot of the fully differenced transformed series from a SEATS adjustment from a seas object generated by the seasonal package.

Usage

```
plot_fully_differenced_transformed_all(
  seas_obj = NULL,
  overall_title = NULL,
```

```

    this_y_label = NULL,
    y_limit = NULL,
    this_x_label = "Time",
    start_plot = NULL,
    do_grid = FALSE,
    do_background = FALSE,
    draw_recess = FALSE,
    recess_color = "lightgrey",
    recess_sub = TRUE,
    title_size = 10,
    add_zero_line = FALSE,
    line_color = "steelblue"
  )

```

Arguments

<code>seas_obj</code>	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
<code>overall_title</code>	Character string; main title of plot. A title will be generated if no title is specified. If <code>this_series = "all"</code> , this title will be the overall plot title and descriptive text will be added.
<code>this_y_label</code>	Character string; y-axis label for plot, if specified.
<code>y_limit</code>	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the series specified.
<code>this_x_label</code>	Label for X axis. Default is "Time"; not displayed if <code>this_series = "all"</code> .
<code>start_plot</code>	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
<code>do_grid</code>	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
<code>do_background</code>	Logical scalar; indicates grey background included in plot. Default is no grey background;
<code>draw_recess</code>	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates. Default is no recession shading.
<code>recess_color</code>	Character string; color used for shading of recession region. Default is 'lightgrey'. For a listing of R color codes, see R Coder (2025).
<code>recess_sub</code>	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label.
<code>title_size</code>	integer scalar; Size of the plot title. If <code>this_series = "all"</code> , the overall title size will be <code>title_size + 2</code> . Default is 10.
<code>add_zero_line</code>	Logical scalar; if TRUE, adds a grey zero line to plots. Default is FALSE, no zero line is added.
<code>line_color</code>	Character string; color used for series in the plot. Default is 'steelblue'. For a listing of R color codes, see R Coder (2025).

Details

Version 2.5.1, 2026-06-09

Value

A ggplot object that generates a plot of the fully differenced transformed series from a SEATS seasonal adjustment. If SEATS seasonal adjustment not produced, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
shoes_seats_seas <-
  seasonal::seas(shoes2008, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    seats.finite = "yes",
    seats.save = c( 'dor', 'dsa', 'dtr' ) )
p_all_full_diff <-
  plot_fully_differenced_transformed_all(shoes_seats_seas,
    overall_title = "US Shoe Sales",
    title_size = 8,
    add_zero_line = TRUE,
    line_color = "darkgreen")
```

plot_irregular_boxplot

Irregular Box Plot from X-13ARIMA-SEATS seasonal adjustment run.

Description

Generate boxplot of irregular component from a seas object generated by the seasonal package.

Usage

```
plot_irregular_boxplot(
  seas_obj = NULL,
  this_table = NULL,
  main_title = NULL,
  sub_title = NULL,
  this_y_label = NULL,
  y_limit = NULL,
  this_x_label = NULL,
  add_notch = FALSE,
```

```

    add_jitter = FALSE,
    jitter_shape = 16,
    jitter_degree = 0.2,
    do_grid = FALSE,
    do_background = FALSE,
    line_color = "black"
  )

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_table	Character string; X-13ARIMA-SEATS table name or abbreviation for an irregular. Possible tables are "irregular", "irregularoutlieradj", "modirregular", "d13", "e3", "s13", "se3" If not a valid table name, the function will print an error message and return a NULL.
main_title	Character string; main title of plot. A title will be generated if no title is specified.
sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot, if specified.
y_limit	Numeric vector of length 2; Range of values for the y-axis. Default is range of the series specified.
this_x_label	Label for X axis. Default is "Month" or "Quarter".
add_notch	Logical scalar; indicates if notches will appear in the box plots around the median. Default is no notches.
add_jitter	Logical scalar; indicates if jittered dots are added to the boxplot. Default is no jittered dots.
jitter_shape	Integer scalar; code for point type used for jittered points. Default is 16.
jitter_degree	Numeric scalar; degree of jittering for points. Default is 0.2
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character string; color used for series in the plot. Default is 'black'. For a listing of R color codes, see R Coder (2025).

Details

Version 1.8.1, 2026-06-09

Value

A ggplot object that generates a boxplot of user-specified irregular series from an X-13ARIMA-SEATS table. If a suitable series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
shoes_seas <-
  seasonal::seas(shoes2008, x11.save = "d13",
    x11 = "", transform.function = "log",
    check.print = c("none", "+acf",
      "+acfplot", "+normalitytest"),
    regression.aictest = c("td", "easter"),
    regression.save = c("td", "hol"),
    outlier.types = "all",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 60)

p_bp1 <-
  plot_irregular_boxplot(shoes_seas, "d13",
    main_title = "X-11 Irregular by Month",
    sub_title = "US Shoe Sales",
    line_color = "steelblue")

p_bp1_notch <-
  plot_irregular_boxplot(shoes_seas, "d13",
    main_title = "X-11 Irregular by Month",
    sub_title = "US Shoe Sales",
    line_color = "steelblue",
    add_notch = TRUE)

ukgas_seats_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    check.print = c( 'pacf', 'pacfplot' ), seats.save = "se3")

p_bp2 <-
  plot_irregular_boxplot(ukgas_seats_seas, "se3",
    main_title = "SEATS Outlier Adj Irregular by Quarter",
    sub_title = "UK Gas Production",
    line_color = "steelblue")

p_bp2_jitter <-
  plot_irregular_boxplot(ukgas_seats_seas, "se3",
    main_title = "SEATS Outlier Adj Irregular by Quarter",
    sub_title = "UK Gas Production",
    line_color = "steelblue",
    add_jitter = TRUE,
    jitter_degree = 0.25)
```

plot_matrix	<i>Plot time series matrix</i>
-------------	--------------------------------

Description

Generate plot of a matrix of user-specified time series.

Usage

```
plot_matrix(
  this_matrix = NULL,
  main_title = deparse(substitute(this_matrix)),
  sub_title = NULL,
  this_y_label = NULL,
  y_limit = NULL,
  this_x_label = "Time",
  start_plot = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  line_color = NULL,
  this_palette = "Dark2",
  this_line_type = "solid",
  this_legend_title = "Series",
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10,
  do_facet = FALSE,
  reset_facet_y_axis = FALSE
)
```

Arguments

this_matrix	Numeric matrix; columns of time series object to be plotted.
main_title	Character string; main title of plot. The default title is the name of the matrix passed to this function.
sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot, if specified.
y_limit	Numeric vector of length 2; Range of values on plot y-axis Default is range of the series specified.
this_x_label	Label for X axis. Default is "Time".
start_plot	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
do_grid	Logical scalar; indicates if plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character scalar; color used for plot. User should specify one color for each column of the matrix specified. Default is the RColorBrewer palette "Dark2". For a listing of R color codes, see R Coder (2025).

`this_palette` Character string; default RColorBrewer palette (Neuwirth 2022). Deault is "Dark2".

`this_line_type` Character string; indicates line type of each plot produced. Default is "solid".

`this_legend_title` Title for legend. Default is "Forecast"

`this_legend_title_size` integer scalar; Size of the legend title. Default is 12.

`this_legend_text_size` integer scalar; Size of the legend title. Default is 10.

`this_legend_position` character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.

`this_legend_space` integer scalar; Space between legend and plot in points. Default is 10.

`do_facet` Logical scalar; indicates if a facet plot is generated of the different colums. Default is FALSE.

`reset_facet_y_axis` Logical scalar; indicates if y-axis for facet plots reset to `y_limit`. Default is FALSE.

Details

Version 6.3, 2026-02-17

Value

A ggplot object that produces a plot of user-specified time series in columns of `this_matrix`, either all on the same axis or as a facet plot. This function uses the `autoplot` function of the `forecast` package (Hyndman and Khandakar 2008; Hyndman et al. 2025). If matrix not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Hyndman R, Athanasopoulos G, Bergmeir C, Caceres G, Chhay L, O'Hara-Wild M, Petropoulos F, Razbash S, Wang E, Yasmeen F (2025). *forecast: Forecasting functions for time series and linear models*. R package version 8.24.0, <https://pkg.robjhyndman.com/forecast/>.

Hyndman RJ, Khandakar Y (2008). "Automatic time series forecasting: the forecast package for R." *Journal of Statistical Software*, **27**(3), 1–22. doi:10.18637/jss.v027.i03.

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
BP_Region_Matrix <-
  cbind(blsplotGG::xt_data_list$mw1u, blsplotGG::xt_data_list$ne1u,
        blsplotGG::xt_data_list$so1u, blsplotGG::xt_data_list$we1u)
colnames(BP_Region_Matrix) <- names(blsplotGG::xt_data_list)
p_BP <- blsplotGG::plot_matrix(BP_Region_Matrix,
  this_y_label = 'Building Permits',
  main_title = "US Building Permits, 1 Family Units",
  do_grid = TRUE, this_line_type = rep("solid", 4),
  line_color = c("orange", "steelblue", "forestgreen", "brown"))
p_BP_facet <- blsplotGG::plot_matrix(BP_Region_Matrix,
  this_y_label = 'Building Permits',
  main_title = "US Building Permits, 1 Family Units",
  do_grid = FALSE, do_facet = TRUE,
  line_color = c("orange", "steelblue", "forestgreen", "brown"))
```

plot_maximum_percent_difference

Maximum percent difference plot

Description

Generates a ggplot object with a time series of the maximum percent difference from a sliding spans analysis of seasonal factors or changes. For more information on sliding spans analysis, see Sections 6.2 and 7.13 of U. S. Census Bureau (2025), with more details on sliding spans analysis in Findley et al. (1990).

Usage

```
plot_maximum_percent_difference(
  seas_obj = NULL,
  this_series = "sfs",
  main_title = "Maximum Percent Difference Plot",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = " ",
  do_grid = FALSE,
  do_background = FALSE,
  line_color = "steelblue",
  cut_color = "red",
  zero_color = "lightgrey"
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; three character code for the type of series to be generated. Allowed entries are "sfs" (seasonal factors, default), "chs" (period-to-period changes), "sis" (indirect seasonal factors), "cis" (indirect period-to-period changes).

main_title	Title for the plot. Default is character string 'Maximum Percent Difference Plot'.
sub_title	Subtitle for the plot. Optional entry.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Color used for lines in the maximum percentage difference plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).
cut_color	Color use to show the slidings spans cut off for this type of series. Default is "red".
zero_color	Color use to show zero line. Default is "lightgrey".

Details

Version 3.0.1, 2026-06-09

Value

A ggplot object that generates a plot of the maximum percent difference from a sliding spans analysis of seasonal factors or changes.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Findley DF, Monsell BC, Shulman HB, Pugh MG (1990). "Sliding Spans Diagnostics for Seasonal and Related Adjustments." *Journal of the American Statistical Association*, **85**, 345–355. doi:10.2307/2289770.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_sspan_seas <-
  seasonal::seas(blsplotGG::shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    slidingspans.save = c('sfs', 'chs'))
```

```
p_shoes_maxpct_sf <-
  plot_maximum_percent_difference(shoes_sspan_seas, "sfs",
    main_title = "Maximum Percent Difference Plot of Seasonal Factors",
    sub_title = "US Shoe Sales",
    line_color = "blue",
    cut_color = "purple",
    zero_color = "black")
```

```
plot_maximum_percent_difference_by_period
```

Maximum percent difference for sliding spans by period

Description

Generate a plot of the maximum percent difference from a sliding spans analysis by month or quarter. For more information on sliding spans analysis, see Sections 6.2 and 7.13 of U. S. Census Bureau (2025), with more details on sliding spans analysis in Findley et al. (1990).

Usage

```
plot_maximum_percent_difference_by_period(
  seas_obj = NULL,
  this_series = "sfs",
  main_title = NULL,
  sub_title = NULL,
  this_y_label = NULL,
  this_x_label = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  point_color = NULL,
  this_palette = "Set3",
  cut_color = "red",
  this_legend_title = "Year",
  this_legend_position = "right",
  this_legend_size = NULL,
  this_legend_order = NULL,
  this_legend_space = NULL
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; three character code for the type of series to be generated. Allowed entries are "sfs" (seasonal factors, default), "chs" (period-to-period changes), "sis" (indirect seasonal factors), "cis" (indirect period-to-period changes).
main_title	Character string; main title of plot. Default is character string 'Maximum Percent Difference Plot by Period', where Period is replaced by Month or Quarter.
sub_title	Character string; subtitle of plot. There is no default subtitle.

this_y_label	Character string; y-axis label for plot, if specified.
this_x_label	Label for X axis. Default is "Month" or "Quarter".
do_grid	Logical scalar; indicates if plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background.
point_color	Character scalar; color used for points in this plot. User should specify one color for each year of the sliding spans. Default is the RColorBrewer palette "Set3". For a listing of R color codes, see R Coder (2025).
this_palette	Character string; default RColorBrewer palette (Neuwirth 2022). Default is "Set3".
cut_color	Color use to show the slidings spans cut off for this type of series. Default is "red".
this_legend_title	Title for legend. Default is "Year".
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_size	Numeric scalar; indicates the size of several attributes of the plot legend, in centimeters. Default value is 1.
this_legend_order	Character vector; indicates the order of entries in the legend. Default order is determined by ggplot2.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 4.6.1, 2026-06-09

Value

Generate ggplot object generating a plot of the maximum percent difference from a sliding spans analysis by month or quarter

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Findley DF, Monsell BC, Shulman HB, Pugh MG (1990). "Sliding Spans Diagnostics for Seasonal and Related Adjustments." *Journal of the American Statistical Association*, **85**, 345–355. doi:10.2307/2289770.

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_sspan_seas <-
  seasonal::seas(blsplotGG::shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    slidingspans.save = c('sfs', 'chs'))
p_shoes_maxpct_sf <-
  plot_maximum_percent_difference_by_period(shoes_sspan_seas, "sfs",
    main_title = "Maximum Percent Difference Plot of Seasonal Factors",
    sub_title = "US Shoe Sales",
    this_palette = "Paired")
```

plot_ratio

Ratio plot

Description

Generates a high-definition plot around a reference line other than zero.

Usage

```
plot_ratio(
  ratio_series = NULL,
  ratio_range = NULL,
  main_title = "Ratio Plot",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = "Ratio",
  do_grid = FALSE,
  do_background = FALSE,
  draw_recess = FALSE,
  recess_color = "lightgrey",
  recess_sub = TRUE,
  add_line = TRUE,
  ratio_mean = 1,
  ratio_color = "steelblue"
)
```

Arguments

ratio_series	Time series of ratios/factors for which you want to generate a high definition plot
--------------	---

ratio_range	Range of values you wish the plot to be plotted over. Default is range of the series.
main_title	Title for the plot. Default is character string 'Ratio Plot'.
sub_title	Subtitle for the plot. Default is NULL.
this_x_label	Label for X axis. Default is "Time".
this_y_label	Label for Y axis. Default is "Ratio".
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates. Default is no recession shading.
recess_color	Character string; color used for shading of recession region. Default is 'lightgrey'. For a listing of R color codes, see R Coder (2025).
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label.
add_line	Logical scalar; add solid line for assumed mean. Default is TRUE.
ratio_mean	Assumed mean value for the ratio. Default is 1.0
ratio_color	Color used for lines in ratio plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).

Details

Version 3.4.1, 2026-06-09

Value

A ggplot object that produces a ratio plot of a time series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    series.appendfcst = "no", x11.seasonalma = "s3x5",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_x11_sf <- seasonal::series(ukgas_x11_seas, "d10")
ukgas_sf_ratio <- plot_ratio(ukgas_x11_sf, main_title = "UK Gas X-11 Seasonal Factors",
  sub_title = "Seasonal Filter: 3x5")
```

plot_ratio_facet	<i>Ratio facet plot</i>
------------------	-------------------------

Description

Generates a high-definition plot for a number of X-13 factors around a reference line other than zero.

Usage

```
plot_ratio_facet(
  seas_obj = NULL,
  ratio_tables = NULL,
  main_title = "Ratio Facet Plot",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = "Ratio",
  do_grid = FALSE,
  do_background = FALSE,
  add_line = TRUE,
  this_scale = "fixed",
  ratio_mean = 1,
  remove_fcst = FALSE,
  ratio_color = "steelblue"
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
ratio_tables	Array of tables for which you want to generate a high definition plot. Possible entries are "sf" (seasonal factor), "td" (trading day factor), "hol" (holiday factors), and "irr" (irregular).
main_title	Title for the plot. Default is character string 'Ratio Facet Plot'.
sub_title	Subtitle for the plot. Default is NULL.
this_x_label	Label for X axis. Default is "Time".
this_y_label	Label for Y axis. Default is "Ratio".
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
add_line	Logical scalar; add solid line for assumed mean. Default is TRUE.
this_scale	Character scalar; set the scale for the y-axis of the facet plots. Default is "fixed". To set the y-axis to be different for the different plots, set this_scale = "free_y".
ratio_mean	Assumed mean value for the ratio. Default is 1.0
remove_fcst	Logical scalar; if TRUE, indicates forecasts to be removed from the series before plotting. Default is no forecasts removed (FALSE);
ratio_color	Color used for lines in ratio plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).

Details

Version 4.3.1, 2026-06-09

Value

A ggplot object that generates a ratio facet plot for the factors provided in the `ratio_tables` argument.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
shoes_x11_seas <-
  seasonal::seas(shoes2008, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 36,
    regression.aictest = c("td", "easter"),
    regression.save = c("td", "hol"),
    x11.seasonalma = "s3x5", check.print = c( 'pacf', 'pacfplot' ))
shoes_factor_ratio_one <-
  plot_ratio_facet(shoes_x11_seas, c("sf", "irr"),
    main_title = "US Retail Shoe Sales",
    sub_title = "X-11 Seasonal (3x5) and Irregular Factors")
shoes_factor_ratio_two <-
  plot_ratio_facet(shoes_x11_seas, c("td", "hol", "irr"),
    main_title = "US Retail Shoe Sales",
    sub_title = "Trading Day, Easter, and Irregular Factors",
    this_scale = "free_y")
```

plot_ratio_two

Ratio plots for two series

Description

Generates ratio plots for two series, and a third plot of the ratio/difference of the two series if the user requests it.

Usage

```
plot_ratio_two(
  ratio_one = NULL,
  ratio_two = NULL,
  ratio_range_one = NULL,
```

```

ratio_range_two = NULL,
do_comparison_plot = TRUE,
overall_title = NULL,
ratio_one_title = "First Ratio Plot",
ratio_two_title = "Second Ratio Plot",
title_size = 10,
remove_fcst_one = NULL,
remove_fcst_two = NULL,
comparison_title = "Comparison Plot",
do_grid = FALSE,
do_background = FALSE,
add_line = TRUE,
ratio_mean_one = NULL,
ratio_mean_two = NULL,
ratio_color = "steelblue"
)

```

Arguments

ratio_one	First time series of ratios/factors for which you want to generate a high definition plot. This is a required entry.
ratio_two	Second time series of ratios/factors for which you want to generate a high definition plot. This is a required entry.
ratio_range_one	Range of values you wish the plot to be plotted over for the first ratio series. Default is range of the first series.
ratio_range_two	Range of values you wish the plot to be plotted over for the second ratio series. Default is range of the second series.
do_comparison_plot	Logical scalar. If TRUE, a ratio/difference plot of the two factors will be generated from ratio_one and ratio_two, if the series have the same periodicity and are the same type of factor. Default is TRUE.
overall_title	Title for the combined plot. Default is NULL.
ratio_one_title	Title for the first plot. Default is character string 'First Ratio Plot'.
ratio_two_title	Title for the plot. Default is character string 'Second Ratio Plot'.
title_size	integer scalar; Size of the component titles. The overall title size will be title_size + 2. Default is 10.
remove_fcst_one	Integer scalar; indicates forecasts to be removed from ratio_one before plotting. Default is no forecasts removed;
remove_fcst_two	Integer scalar; indicates forecasts to be removed from ratio_two before plotting. Default is no forecasts removed;
comparison_title	Title for the comparison plot, if requested. Default is character string 'Comparison Plot'.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.

do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
add_line	Logical scalar; add solid line for assumed mean. Default is TRUE.
ratio_mean_one	Assumed mean for the first ratio series. Default is 1.0 if ratio_range_one has both elements greater than zero, 0.0 otherwise.
ratio_mean_two	Assumed mean for the second ratio series. Default is 1.0 if ratio_range_two has both elements greater than zero, 0.0 otherwise.
ratio_color	Color used for lines in ratio plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).

Details

Version 3.5.1, 2026-06-09

Value

A ggplot object that generates a stacked plot with the

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    x11.seasonalma = "s3x5", x11.save = c("d10", "d11"),
    check.print = c( 'pacf', 'pacfplot' ), series.appendfcst = "no")
ukgas_seats_seas <-
  seasonal::seas(UKgas, series.period = 4, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    seats.save = c("s10", "s11"), series.appendfcst = "no",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_x11_sf <- seasonal::series(ukgas_x11_seas, "d10")
ukgas_seats_sf <- seasonal::series(ukgas_seats_seas, "s10")
ukgas_sf_two_plot <-
  plot_ratio_two(ukgas_x11_sf, ukgas_seats_sf,
    overall_title = "UK Gas Production",
    ratio_one_title = "X-11 Seasonal Factors",
    ratio_two_title = "SEATS Seasonal Factors",
    comparison_title = "Ratio of Seasonal Factors (X11 / SEATS)",
    ratio_color = "darkgreen")

ukgas_seats_add_seas <-
  seasonal::seas(UKgas, series.period = 4, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "none", forecast.maxlead = 20,
    seats.save = c("s10", "s11"), series.appendfcst = "no",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_seats_add_sf <- seasonal::series(ukgas_seats_add_seas, "s10")
```

```

ukgas_sf_two_plot_add_mult <-
  plot_ratio_two(ukgas_seats_add_sf, ukgas_seats_sf,
    overall_title = "UK Gas Production",
    ratio_one_title = "SEATS Additive Seasonal Factors",
    ratio_two_title = "SEATS Seasonal Factors",
    do_comparison_plot = FALSE,
    ratio_color = "darkgreen")

```

plot_resid	<i>Residual plot</i>
------------	----------------------

Description

Generates a plot of the regARIMA residuals extracted from a seas object, with diagnostic information added to the subtitle.

Usage

```

plot_resid(
  seas_obj = NULL,
  main_title = "ARIMA Residuals",
  series_name = NULL,
  this_x_label = "Time",
  this_y_label = NULL,
  do_grid = TRUE,
  do_background = FALSE,
  draw_recess = FALSE,
  recess_color = NULL,
  recess_sub = TRUE,
  use_ratio = FALSE,
  add_line = TRUE,
  line_color = "steelblue"
)

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
main_title	Character string; main title of plot. Default is 'ARIMA Residuals'.
series_name	Character scalar; name of the time series used in seas_obj.
this_x_label	Label for X axis. Default is "Time".
this_y_label	Label for Y axis. Default is series_name. if specified.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is grid lines plotted.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates. Default is recession shading not plotted.

recess_color	Character string; color used for shading of recession region. Default is 'lightgrey'.
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label is produced
use_ratio	Logical scalar; indicates if plots of seasonal factors, irregular, and residuals are done as ratio plots. Default has these plots as time series line plots.
add_line	Logical scalar; add solid line for assumed mean. Default is TRUE.
line_color	Character string; color used for residuals. Default is "green". For a listing of R color codes, see R Coder (2025).

Details

Version 3.6.1, 2026-06-09

Value

Generates a ggplot object of a plot of the regARIMA residuals with diagnostic information in the sub-headers.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = "(0 1 1)(0 1 1)")
plot_resid_air <-
  plot_resid(air_seas,
    main_title = "ARIMA Residuals for Airline Passengers",
    use_ratio = TRUE,
    line_color="darkblue")
```

plot_sadj_and_ori	<i>Plot X-13 seasonal adjustment, original series on same axis</i>
-------------------	--

Description

Generates a ggplot object with a time series plot that compares an X-13 seasonal adjustment and trend, optionally including the original series.

Usage

```
plot_sadj_and_ori(
  this_sa = NULL,
  this_ori = NULL,
  this_sa_type = "SEATS",
  main_title = NULL,
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = " ",
  do_grid = FALSE,
  do_background = FALSE,
  line_color = c("grey", "darkblue"),
  line_label = c("Ori", "SA"),
  this_guide_legend = "Series"
)
```

Arguments

<code>this_sa</code>	Time series of a seasonal adjustment. This is a required entry.
<code>this_ori</code>	Time series of the original series. Optional entry.
<code>this_sa_type</code>	Character string; type of seasonal adjustment. Default is "SEATS".
<code>main_title</code>	Title for the plot. By default, the routine will generate a title based on the type of adjustment (X-11 and SEATS) done.
<code>sub_title</code>	Subtitle for the plot. Optional entry.
<code>this_x_label</code>	Label for X-axis. Default is "Time"
<code>this_y_label</code>	Label for Y-axis. Default is " "
<code>do_grid</code>	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
<code>do_background</code>	Logical scalar; indicates grey background included in plot. Default is no grey background;
<code>line_color</code>	Character vector of length 2; color used for lines in the plot, in the order of original series, seasonally adjusted series. Default is c("grey", "darkblue"). For a listing of R color codes, see R Coder (2025).
<code>line_label</code>	Character vector of length 2; labels used for lines in the plot, in the order of original series, seasonally adjusted series. Default is c("Ori", "SA").
<code>this_guide_legend</code>	Title for legend. Default is "Series"

Details

Version 2.4, 2026-02-17

Value

A ggplot object that generates a plot comparing a seasonally adjusted series with the original series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 20,
    x11.seasonalma = "s3x5",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_x11_sa <- seasonal::final(ukgas_x11_seas)
ukgas_sa_ori_p <-
  plot_sadj_and_ori(this_sa = ukgas_x11_sa,
    this_ori = UKgas,
    this_sa_type = "X-11",
    main_title = "UK Gas Seasonal Adjustment",
    line_color = c("lightgrey", "forestgreen"))
```

plot_sadj_and_trend *Plot X-13 seasonal adjustment, trend on same axis*

Description

Generates a ggplot object with a time series plot that compares an X-13 seasonal adjustment and trend, extracted from a seas object. The plot optionally includes the original series.

Usage

```
plot_sadj_and_trend(
  seas_obj = NULL,
  plot_ori = TRUE,
  main_title = NULL,
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = " ",
  do_grid = FALSE,
  do_background = FALSE,
  line_color = NULL,
  this_palette = "Dark2",
  line_label = NULL,
  this_legend_title = "Series",
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
plot_ori	Logical scalar, indicates if original series is included with plot. Default is TRUE.
main_title	Title for the plot. By default, the routine will generate a trend based on the type of adjustment (X-11 and SEATS) done.
sub_title	Subtitle for the plot. Optional entry.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character vector of length 2 (if plot_ori is FALSE) or 3 (if plot_ori is TRUE); color used for lines in the plot, in the order of original series (if specified), seasonally adjusted series, trend. Default is generated from the RColorBrewer palette "Dark2". For a listing of R color codes, see R Coder (2025).
this_palette	Character string; default RColorBrewer palette (Neuwirth 2022). Deault is "Dark2".
line_label	Character vector of length 2; labels used for lines in the plot, in the order of original series, seasonally adjusted series. Default is c("Ori", "SA", "Trend") (if original series plotted) or c("SA", "Trend").
this_legend_title	Title for legend. Default is "Series"
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.
this_legend_text_size	integer scalar; Size of the legend title. Default is 10.
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 3.5.1, 2026-06-09

Value

A ggplot object that generates a plot comparing a seasonally adjusted series with the trend generated from the same X-13ARIMA-SEATS seasonal adjustment.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    x11.seasonalma = "s3x5", x11.save = "d12",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_x11_sadj_and_trend_p <-
  plot_sadj_and_trend(ukgas_x11_seas, plot_ori = TRUE,
    main_title = "UK Gas",
    sub_title = "X-11 Seasonal Adjustment",
    line_color = c("lightgrey", "steelblue", "forestgreen"))
ukgas_seats_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    seats.save = "s12",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_seats_sadj_and_trend_p <-
  plot_sadj_and_trend(ukgas_seats_seas, plot_ori = FALSE,
    main_title = "UK Gas Series",
    sub_title = "SEATS Seasonal Adjustment",
    line_color = c("steelblue", "forestgreen"))
```

plot_sadj_and_trend_facet

Plot X-13 seasonal adjustment, trend in a facet plot

Description

Generates a ggplot object with a time series facet plot that compares an X-13 seasonal adjustment and trend, extracted from a seas object. The plot optionally includes the original series.

Usage

```
plot_sadj_and_trend_facet(
  seas_obj = NULL,
  plot_ori = TRUE,
  main_title = NULL,
  sub_title = NULL,
  this_x_label = "Time",
```

```

    this_y_label = " ",
    do_grid = FALSE,
    do_background = FALSE,
    line_color = NULL,
    this_palette = "Dark2",
    line_label = NULL,
    this_scale = "fixed"
  )

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
plot_ori	Logical scalar, indicates if original series is included with plot. Default is TRUE.
main_title	Character string; title for the plot. By default, the routine will generate a title based on the type of adjustment (X-11 and SEATS) done.
sub_title	Subtitle for the plot. Optional entry.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character vector of length 2 (if plot_ori is FALSE) or 3 (if plot_ori is TRUE); color used for lines in the plot, in the order of original series (if specified), seasonally adjusted series, trend. Default is generated from the RColorBrewer palette "Dark2". For a listing of R color codes, see R Coder (2025).
this_palette	Character string; default RColorBrewer palette (Neuwirth 2022). Default is "Dark2".
line_label	Character vector of length 2; labels used for lines in the plot, in the order of original series, seasonally adjusted series. Default is c("Ori", "SA", "Trend") (if original series plotted) or c("SA", "Trend").
this_scale	Character scalar; set the scale for the y-axis of the facet plots. Default is "fixed". To set axis to be different for the different plots, set this_scale = "free_y".

Details

Version 6.3.1, 2026-06-09

Value

A ggplot object that generates a facet plot comparing a seasonally adjusted series with the trend generated from the same X-13ARIMA-SEATS seasonal adjustment.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    x11.seasonalma = "s3x5", x11.save = "d12",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_x11_sadj_and_trend_facet_p <-
  plot_sadj_and_trend_facet(ukgas_x11_seas, plot_ori = TRUE,
    main_title = "UK Gas",
    sub_title = "X-11 Seasonal Adjustment",
    line_color = c("steelblue", "forestgreen", "grey"))
ukgas_seats_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    seats.save = "s12",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_seats_sadj_and_trend_facet_p <-
  plot_sadj_and_trend_facet(ukgas_seats_seas, plot_ori = FALSE,
    main_title = "UK Gas Series",
    sub_title = "SEATS Seasonal Adjustment",
    line_color = c("steelblue", "forestgreen"))
```

plot_sa_history

Revisions History Plot for Seasonal Adjustments

Description

Generates a ggplot2 object of estimates from a revisions history of a seasonal adjustment for a given series. For more information on revisions history diagnostics, see Sections 6.3 and 7.8 of U. S. Census Bureau (2025).

Usage

```
plot_sa_history(
  seas_obj = NULL,
  add_ori = TRUE,
  main_title = "Seasonal Adjustment History Graph",
  sub_title = NULL,
  this_x_label = "Time",
```

```

    this_y_label = NULL,
    do_grid = FALSE,
    do_background = FALSE,
    line_color = NULL,
    this_palette = "Dark2"
  )

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
add_ori	Logical scalar; add the original series to the plot. Default is TRUE.
main_title	Character string. Title for the plot. Default is 'Seasonal Adjustment History Graph'.
sub_title	Subtitle for the plot. Default is NULL.
this_x_label	Label for X axis. Default is "Time".
this_y_label	Label for Y axis. Default is NULL.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Vector with colors used for lines in history plot. Should be of length 4 (if add_ori = FALSE) or 5 (if add_ori = TRUE) Default is NULL, which indicates that the palette specified in this_palette is used to generate colors for this plot. For a listing of R color codes, see R Coder (2025).
this_palette	Colors used for lines in plot. Default is RColorBrewer palette "Dark2" (Neuwirth 2022).

Details

Version 2.3.1, 2026-06-09

Value

A ggplot object that generates a history plot of the seasonal adjustment. The seas_obj argument must contain output for a revisions history analysis for seasonal adjustments with sadjlags set to 1 and 12 (for monthly series) or 4 (for quarterly series).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of*

Statistical Software, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_seas <-
  seasonal::seas(shoes2008, x11.save = "d13",
    x11 = "", transform.function = "log",
    check.print = c("none", "+acf", "+acfplot", "+normalitytest"),
    regression.aictest = c("td", "easter"),
    regression.save = c("td", "hol"),
    outlier.types = "all",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 60,
    history.estimates = c("sadj", "sadjchng", "trend"),
    history.sadjlags = c(1,12),
    history.print = "all",
    history.save = c("sae", "sar"))
p_shoes_sa_history <-
  plot_sa_history(shoes_seas, add_ori = FALSE,
    main_title = "SA History Graph, Lag 1 and 12",
    sub_title = "US Retail Sales of Shoes")
```

plot_seasonal_sums	<i>Plot of the seasonal period length sums of the SEATS seasonal factors from an X-13ARIMA-SEATS SEATS seasonal adjustment run.</i>
--------------------	---

Description

Generate plot of the seasonal period length sums of the SEATS seasonal factors from a SEATS adjustment from a seas object generated by the seasonal package. For more information on SEATS input options in X-13ARIMA-SEATS, (see Section 7.14 of U. S. Census Bureau 2025).

Usage

```
plot_seasonal_sums(
  seas_obj = NULL,
  main_title = NULL,
  sub_title = NULL,
  this_y_label = NULL,
  this_x_label = "Time",
  do_grid = FALSE,
  do_background = FALSE,
  draw_recess = FALSE,
  recess_color = "lightgrey",
  recess_sub = TRUE,
  line_color = "steelblue"
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
main_title	Character string; main title of plot. A title will be generated if no title is specified.
sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot, if specified.
this_x_label	Label for X axis. Default is "Time".
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates. Default is no recession shading.
recess_color	Character string; color used for shading of recession region. Default is 'lightgrey'.
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label.
line_color	Character string; color used for series in the plot. Default is 'steelblue'. For a listing of R color codes, see R Coder (2025).

Details

Version 1.8.1, 2026-06-09

Value

A ggplot object which generates a plot of the seasonal period length sums of the SEATS seasonal factors. If SEATS seasonal adjustment not produced, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_seats_seas <-
  seasonal::seas(shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    seats.finite = "yes",
    seats.save = c( 'ssm' ) )
p_seasonal_sum <-
  plot_seasonal_sums(shoes_seats_seas,
    sub_title = "US Shoe Sales",
    do_grid = TRUE,
    line_color = "darkgreen")
```

plot_seats_filter	<i>Plot adjustment or trend filter from an X-13ARIMA-SEATS SEATS seasonal adjustment run.</i>
-------------------	---

Description

Generate plot of the fully differenced transformed series from a SEATS adjustment from a seas object generated by the seasonal package. For more information on SEATS input options in X-13ARIMA-SEATS, (see Section 7.14 of U. S. Census Bureau 2025).

Usage

```
plot_seats_filter(
  seas_obj = NULL,
  this_series = "sa",
  this_filter = "symmetric",
  main_title = NULL,
  sub_title = NULL,
  this_y_label = "Filter Coefficient",
  this_x_label = "Index",
  this_x_breaks = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  title_size = 12,
  point_color = "steelblue",
  point_size = 1.5,
  point_shape = 20
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; series for which SEATS produces a set of filter weights, limited to the seasonally adjusted series (sa, the default), or the trend component (trend). For other entries, the function will print an error message and return a NULL.

this_filter	Character string; type of filter for which SEATS produces filter coefficients, limited to the symmetric (symmetric, the default), or the concurrent (trend) filter. For other entries, the function will print an error message and return a NULL.
main_title	Character string; main title of plot. A title will be generated if no title is specified.
sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot. Default is "Filter Coefficient".
this_x_label	Label for X axis. Default is "Index".
this_x_breaks	Numeric array with tick marks for the X-axis. Default is generated by the plotting routine.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
title_size	integer scalar; Size of the plot title. Default is 12.
point_color	Character string; color used for points in the plot. Default is 'steelblue'. For a listing of R color codes, see R Coder (2025).
point_size	Integer scalar; relative size of points in filter plot. Default is 1.5.
point_shape	Integer scalar; code for the shape of points in filter plot. Default is 20, a small filled circle.

Details

Version 3.3.1, 2026-06-09

Value

A ggplot object that generates a plot of the adjustment or trend filter from a SEATS seasonal adjustment from X-13ARIMA-SEATS. If SEATS seasonal adjustment (with `seats.finite = "yes"`) not produced, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

- R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].
- Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.
- U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_seats_seas <-
  seasonal::seas(shoes2008, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    seats.finite = "yes",
    seats.save = c( 'fac', 'faf', 'ftc', 'ftf' ) )

p_sa_sym_filter <-
  plot_seats_filter(shoes_seats_seas, "sa", "symmetric",
    sub_title = "US Shoe Sales",
    point_color = "darkgreen",
    this_x_breaks = seq(-96,96,24))

p_trn_sym_filter <-
  plot_seats_filter(shoes_seats_seas, "trn", "symmetric",
    sub_title = "US Shoe Sales",
    point_color = "darkgreen",
    this_x_breaks = seq(-96,96,24))

p_sa_conc_filter <-
  plot_seats_filter(shoes_seats_seas, "sa", "concurrent",
    sub_title = "US Shoe Sales",
    point_color = "darkgreen")

p_trn_conc_filter <-
  plot_seats_filter(shoes_seats_seas, "trn", "concurrent",
    sub_title = "US Shoe Sales",
    point_color = "darkgreen")
```

`plot_seats_filter_all` *Plots adjustment and trend filters from an X-13ARIMA-SEATS SEATS seasonal adjustment run in a single plot.*

Description

Generate plot of the concurrent and symmetric filters for the seasonally adjusted generated by the seasonal package. For more information on SEATS as implemented in X-13ARIMA-SEATS, (see Section 7.14 of U. S. Census Bureau 2025).

Usage

```
plot_seats_filter_all(
  seas_obj = NULL,
  overall_title = "SEATS Adjustment and Trend Filters",
  this_y_label = NULL,
  this_x_label = "Index",
  this_x_concurrent_breaks = NULL,
  this_x_symmetric_breaks = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  title_size = 10,
  point_color = "steelblue",
  point_size = 1.25,
  point_shape = 20
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
overall_title	Character string; main title of plot. Default is "SEATS Adjustment and Trend Filters".
this_y_label	Character string; y-axis label for plot. Default is no y axis label.
this_x_label	Label for X axis. Default is "Index".
this_x_concurrent_breaks	Numeric array with tick marks for the X-axis for concurrent filter plots. Default is generated by the plotting routine.
this_x_symmetric_breaks	Numeric array with tick marks for the X-axis for symmetric filter plots. Default is generated by the plotting routine.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
title_size	integer scalar; Size of the plot title for individual panels. Default is 10; the overall title size will be title_size + 10.
point_color	Character string; color used for points in the plot. Default is 'steelblue'. For a listing of R color codes, see R Coder (2025).
point_size	Integer scalar; relative size of points in filter plot. Default is 1.0.
point_shape	Integer scalar; code for the shape of points in filter plot. Default is 20, a small filled circle.

Details

Version 1.10.1, 2026-06-09

Value

A ggplot object that generates a plot of the adjustment and trend concurrent and symmetric filters from a SEATS seasonal adjustment from X-13ARIMA-SEATS. If SEATS seasonal adjustment (with seats.finite = "yes") not produced, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```

shoes_seats_seas <-
  seasonal::seas(shoes2008, arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    seats.finite = "yes",
    seats.save = c( 'fac', 'faf', 'ftc', 'ftf' ) )
p_shoe_seats_filter <-
  plot_seats_filter_all(shoes_seats_seas,
    overall_title = "US Shoe Sales",
    point_color = "darkgreen",
    this_x_symmetric_breaks = seq(-96,96,24))

```

plot_series	<i>Plot time series object.</i>
-------------	---------------------------------

Description

Generate plot of user-specified time series (ts) object.

Usage

```

plot_series(
  this_series = NULL,
  plot_range = NULL,
  this_series_name = NULL,
  main_title = NULL,
  sub_title = NULL,
  this_y_label = NULL,
  y_limit = NULL,
  this_x_label = "Time",
  start_plot = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  draw_recess = FALSE,
  recess_color = "lightgrey",
  recess_sub = TRUE,
  this_line_type = "solid",
  line_color = "grey"
)

```

Arguments

this_series	A ts object; This is a required entry.
plot_range	Range of values you wish the plot to be plotted over. Default is range of the series.
this_series_name	Character string; name of time series. No default.
main_title	Character string; main title of plot. A title will be generated if no title is specified.

sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot. If not specified, set to this_series_name, if specified.
y_limit	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the series specified.
this_x_label	Label for X axis. Default is "Time".
start_plot	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates. Default is no recession shading.
recess_color	Character string; color used for shading of recession region. Default is 'lightgrey'. For a listing of R color codes, see R Coder (2025).
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label.
this_line_type	Character string; indicates line type of each plot produced. Default is "solid".
line_color	Character string; color used for series in the plot. Default is 'grey'. For a listing of R color codes, see R Coder (2025).

Details

Version 3.0.2, 2026-06-09

Value

Generate ggplot plot of user-specified series. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11="",
    transform.function = "log")
air_seas_d11 <- seasonal::final(air_seas)
air_d11_p <-
  plot_series(air_seas_d11,
    this_series_name = "AirPassengers",
    main_title = 'X-11 Seasonal Adjustment of Airline Passengers',
    sub_title = 'Box-Jenkins Airline series',
```

```
do_grid = TRUE,
draw_recess = TRUE,
line_color = "darkblue")
```

plot_sf_heat_map	<i>Seasonal factor heat map</i>
------------------	---------------------------------

Description

Generates a heat map of the seasonal factors.

Usage

```
plot_sf_heat_map(
  this_sf = NULL,
  main_title = "Seasonal Heat Map",
  sub_title = NULL,
  this_x_label = NULL,
  this_y_label = NULL,
  this_palette = "YlOrRd"
)
```

Arguments

this_sf	array of seasonal factors stored as a time series
main_title	Character string; main title of plot. Default is 'Seasonal Sub-Plots'.
sub_title	Character string; subtitle of plot. Subtitle not produced if not specified.
this_x_label	Character string; label for x-axis of plot. Default is a blank x-axis.
this_y_label	Character string; label for y-axis of plot. Default is a blank y-axis.
this_palette	Character string; default RColorBrewer palette (Neuwirth 2022). Deault is "YlOrRd".

Details

Version 1.2, 2026-07-21

Value

A ggplot object which generates a heat map of the seasonal factors.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

Examples

```

air_seas <-
  seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)',
                 x11 = '')
air_sf <- seasonal::series(air_seas, 'd10')

air_sf_heat_map <-
  plot_sf_heat_map(air_sf,
    main_title = "Airline Passengers Heat Map",
    sub_title = "X-11 Seasonal Factors",
    this_x_label = "Month",
    this_palette = "YlGnBu")

```

plot_sf_mean	<i>Seasonal factor mean plot using ggplot</i>
--------------	---

Description

Generates a plot of the means of a matrix of seasonal factors. Each column of the matrix should be a ts object.

Usage

```

plot_sf_mean(
  this_sf_matrix = NULL,
  main_title = deparse(substitute(this_sf_matrix)),
  sub_title = NULL,
  this_y_label = NULL,
  this_x_label = "Time",
  do_grid = FALSE,
  do_background = FALSE,
  line_color = NULL,
  this_palette = "Dark2",
  this_line_type = rep("solid", ncol(this_sf_matrix)),
  do_facet = FALSE,
  this_scale = "free_y",
  y_limit = NULL,
  forecast = 0,
  this_legend_entry = colnames(this_sf_matrix),
  this_legend_title = "SF Means",
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)

```

Arguments

this_sf_matrix time series matrix object of the seasonal factors from a number of seasonal adjustments. This is a required entry.

main_title Character string; main title of plot. Default is 'Mean of Seasonal Factors'.

sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot, if specified.
this_x_label	Label for X axis. Default is "Time".
do_grid	Logical scalar; indicates if plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character scalar; color used for plot. User should specify one color for each column of the matrix specified. For a listing of R color codes, see R Coder (2025). Default is the RColorBrewer palette "Dark2" (Neuwirth 2022).
this_palette	Character string; default RColorBrewer palette Deault is "Dark2".
this_line_type	Character string; indicates line type of each plot produced. Default is rep("solid", ncol(this_sf_matrix)).
do_facet	Logical scalar; indicates if a facet plot is generated of the different columes. Default is FALSE.
this_scale	Character scalar; set the scale for the y-axis of the facet plots. Default is "free_y". To set axis to be the same for the different plots, set this_scale = "fixed".
y_limit	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the seasonal factors.
forecast	Integer scalar; Number of forecasts appended to the seasonal factors. Default is 0.
this_legend_entry	Character array; entries for the legend. When do_facet = TRUE, entries are used as the facet labels. Default is colnames(this_sf_matrix)
this_legend_title	Character string; indicates title of legend. Default is 'Series'.
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.
this_legend_text_size	integer scalar; Size of the legend title. Default is 10.
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 5.6, 2026-02-20

Value

Generate plot of the means of seasonal factors by period. If seasonal factors not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
EM_individual_seas <-
  seasonal::seas(employment_list,
    x11 = "", transform.function = "log",
    check.print = c("none", "+acf", "+acfplot", "+normalitytest"),
    regression.aictest = NULL,
    outlier.types = "all",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 60,
  )
# Use Filter function to grab seas objects
EM_individual_seas_only <-
  Filter(function(x) inherits(x, "seas"), EM_individual_seas)

EM_names <- names(employment_list)
names(EM_individual_seas_only) <- EM_names

EM_Comp_Sf <-
  cbind(seasonal::series(EM_individual_seas_only$`2000013`, "d10"),
    seasonal::series(EM_individual_seas_only$`2000014`, "d10"),
    seasonal::series(EM_individual_seas_only$`2000025`, "d10"),
    seasonal::series(EM_individual_seas_only$`2000026`, "d10"))
colnames(EM_Comp_Sf) <- EM_names

em_plot <- blsplotGG::plot_sf_mean(EM_Comp_Sf,
  main_title = 'US Employment Seasonal Means',
  sub_title = 'X-11 Seasonals',
  forecast = 60,
  this_legend_title = "SF Means",
  this_legend_entry = c("M 16-19", "F 16-19", "M 20+", "F 20+")
)

em_plot_facet <- blsplotGG::plot_sf_mean(EM_Comp_Sf,
  main_title = 'US Employment Seasonal Means',
  sub_title = 'X-11 Seasonals',
  forecast = 60,
  do_facet = TRUE,
  this_scale = "fixed",
  this_legend_entry = c("M 16-19", "F 16-19", "M 20+", "F 20+")
)
```

plot_sf_series

Seasonal factor plot grouped by month/quarter

Description

Generates a special plot of the seasonal factors grouped by month/quarter.

Usage

```
plot_sf_series(
  this_sf = NULL,
  y_limit = NULL,
  this_trans = TRUE,
  main_title = "Seasonal Sub-Plots",
  sub_title = NULL,
  this_xlab = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  line_color_sf = "darkblue",
  line_color_mean = "darkgrey",
  first_year = NULL,
  add_mean_line = TRUE,
  this_legend_title = "SF Plot",
  this_legend_text = c("SF", "SF Mean"),
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

<code>this_sf</code>	array of seasonal factors stored as a time series
<code>y_limit</code>	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the seasonal factors.
<code>this_trans</code>	Logical scalar; indicates if the adjustment was done with a log transform. Default is TRUE.
<code>main_title</code>	Character string; main title of plot. Default is 'Seasonal Sub-Plots'.
<code>sub_title</code>	Character string; subtitle of plot. Subtitle not produced if not specified.
<code>this_xlab</code>	Character string; label for x-axis of plot. Default is a blank x-axis.
<code>do_grid</code>	Logical scalar; indicates if plots will have grid lines. Default is no grid lines.
<code>do_background</code>	Logical scalar; indicates grey background included in plot. Default is no grey background;
<code>line_color_sf</code>	Character string; color used for seasonal factors. Default is "darkblue". For a listing of R color codes, see R Coder (2025).
<code>line_color_mean</code>	Character string; color used for means of the seasonal factors. Default is "darkgrey". For a listing of R color codes, see R Coder (2025).
<code>first_year</code>	Integer scalar; First year used in plot. Default is start of the series.
<code>add_mean_line</code>	Logical scalar; indicates if seasonal factor plots will include lines for seasonal means. Default includes lines for seasonal means.
<code>this_legend_title</code>	Character string; indicates title of legend. Default is 'Series'.
<code>this_legend_text</code>	Array of character strings; indicates text for each seasonal factor in plot. Default is c("SF", "SF Mean").

`this_legend_title_size`
integer scalar; Size of the legend title. Default is 12.

`this_legend_text_size`
integer scalar; Size of the legend title. Default is 10.

`this_legend_position`
character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.

`this_legend_space`
integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 3.2, 2026-06-09

Value

A ggplot object which generates a plot of the seasonal factors (and the SI-ratios) grouped by month/quarter.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)',
    x11 = '')
air_sf <- seasonal::series(air_seas, 'd10')
plot_air_sf <- plot_sf_series(air_sf,
  add_mean_line = TRUE,
  main_title = 'Air Passengers Seasonal Sub-Plots',
  line_color_sf = 'darkgreen',
  line_color_mean = 'lightgreen',
  this_legend_title = 'X-11 Seasonal',
  this_legend_text = c('sf', 'mean'),
  this_legend_position = "bottom",
  this_legend_space = 0)
```

plot_sf_two

Seasonal factor plot grouped by month/quarter for two series

Description

Generates a plot of the seasonal factors grouped by month/quarter for two adjustments.

Usage

```
plot_sf_two(
  this_sf_one = NULL,
  this_sf_two = NULL,
  y_limit = NULL,
  this_trans = TRUE,
  main_title = "Seasonal Sub-Plots",
  sub_title = NULL,
  this_xlab = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  sf_color = NULL,
  mean_color = NULL,
  this_palette = "Paired",
  first_year = NULL,
  this_legend_title = "SF Plot",
  this_legend_text = c("SF One", "SF Mean One", "SF Two", "SF Mean Two"),
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

<code>this_sf_one</code>	array of seasonal factors stored as a time series
<code>this_sf_two</code>	array of seasonal factors stored as a time series
<code>y_limit</code>	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the seasonal factors.
<code>this_trans</code>	Logical scalar; indicates if the adjustment was done with a log transform. Default is TRUE.
<code>main_title</code>	Character string; main title of plot. Default is 'Seasonal Sub-Plots'.
<code>sub_title</code>	Character string; subtitle of plot. Subtitle not produced if not specified.
<code>this_xlab</code>	Character string; label for x-axis of plot. Default is a blank x-axis.
<code>do_grid</code>	Logical scalar; indicates if plots will have grid lines. Default is no grid lines.
<code>do_background</code>	Logical scalar; indicates grey background included in plot. Default is no grey background;
<code>sf_color</code>	Vector of character strings; colors used for seasonal factors. Should be of length two. Default is NULL, which indicates that the palette specified in <code>this_palette</code> is used to generate colors for this plot. For a listing of R color codes, see R Coder (2025).
<code>mean_color</code>	Vector of character strings; color used for means of the seasonal factors. Should be of length two. Default is NULL, which indicates that the palette specified in <code>this_palette</code> is used to generate colors for this plot.
<code>this_palette</code>	Color used for lines in plot. Default is the RColorBrewer palette "Paired" (Neuwirth 2022).
<code>first_year</code>	Integer scalar; First year used in plot. Default is start of the series.
<code>this_legend_title</code>	Character string; indicates title of legend. Default is 'Series'.

`this_legend_text`
 Array of character strings; indicates text for each seasonal factor in plot. Default is `c("SF One", "SF Mean One", "SF Two", "SF Mean Two")`.

`this_legend_title_size`
 integer scalar; Size of the legend title. Default is 12.

`this_legend_text_size`
 integer scalar; Size of the legend title. Default is 10.

`this_legend_position`
 character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.

`this_legend_space`
 integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 2.5, 2026-06-09

Value

A ggplot object which generates a plot of the seasonal factors (and the SI-ratios) grouped by month/quarter.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
air_seas_x11 <-
  seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)',
    x11 = '')
air_sf_x11 <- seasonal::series(air_seas_x11, 'd10')
air_seas_seats <-
  seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)',
    seats.save = "s10")
air_sf_seats <- seasonal::series(air_seas_seats, 's10')
plot_air_sf_two <- plot_sf_two(air_sf_x11, air_sf_seats,
  main_title = 'Air Passengers Seasonal Sub-Plots',
  this_legend_text = c('X-11 sf', 'X-11 mean', 'SEATS sf', 'SEATS mean'),
  this_legend_title_size = 10,
  this_legend_text_size = 8)
```

plot_sliding_spans	<i>Compare spans from the sliding spans analysis</i>
--------------------	--

Description

Generates a ggplot object with a time series plot that compares the series from each of the sliding spans generated from a sliding spans analysis. For more information on sliding spans analysis, see Sections 6.2 and 7.13 of U. S. Census Bureau (2025), with more details in Findley et al. (1990).

Usage

```
plot_sliding_spans(
  seas_obj = NULL,
  this_series = "sfs",
  main_title = "Plot of Sliding Spans",
  sub_title = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  line_color = NULL,
  this_palette = "Dark2",
  this_legend_title = "Sliding Spans",
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; three character code for the type of series to be generated. Allowed entries are "sfs" (seasonal factors, default), "chs" (period-to-period changes), "sis" (indirect seasonal factors), "cis" (indirect period-to-period changes).
main_title	Title for the plot. Default is character string 'Plot of Sliding Spans'.
sub_title	Subtitle for the plot. Optional entry.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character vector of length 2 to 4 based on the number of spans. For a listing of R color codes, see R Coder (2025). Default is the RColorBrewer palette "Dark" (Neuwirth 2022).
this_palette	Character string; default RColorBrewer palette. Deault is "Dark2".
this_legend_title	Title for legend. Default is "Sliding Spans"
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.

`this_legend_text_size`
integer scalar; Size of the legend title. Default is 10.

`this_legend_position`
character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.

`this_legend_space`
integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 2.5.1, 2026-06-09

Value

A ggplot object that generates a plot compares the series from each of the sliding spans generated from a sliding spans analysis.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Findley DF, Monsell BC, Shulman HB, Pugh MG (1990). "Sliding Spans Diagnostics for Seasonal and Related Adjustments." *Journal of the American Statistical Association*, **85**, 345–355. doi:10.2307/2289770.

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_sspan_seas <-
  seasonal::seas(blsplotGG::shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    slidingspans.save = c('sfs', 'chs'))
p_shoes_spans_sf <-
  plot_sliding_spans(shoes_sspan_seas, "sfs",
    main_title = "Spans of Seasonal Factors",
    sub_title = "US Shoe Sales",
    this_palette = "Set1")
```

plot_sliding_spans_two

Maximum percent difference plot for two sliding spans runs

Description

Generates a ggplot object comparing the time series of the maximum percent difference from the sliding spans analysis of seasonal factors or changes for two sliding spans runs. The sliding spans analysis for both series should have the same length of spans. For more information on sliding spans analysis, see Sections 6.2 and 7.13 of U. S. Census Bureau (2025), with more details in Findley et al. (1990).

Usage

```
plot_sliding_spans_two(
  seas_obj_one = NULL,
  seas_obj_two = NULL,
  this_series = "sfs",
  main_title = "Maximum Percent Difference Comparison Plot",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = " ",
  do_grid = FALSE,
  do_background = FALSE,
  cut_color = "red",
  this_legend_title = "Sliding Spans",
  this_legend_entry = c("ss1", "ss2"),
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

seas_obj_one	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
seas_obj_two	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; three character code for the type of series to be generated. Allowed entries are "sfs" (seasonal factors, default), "chs" (period-to-period changes), "sis" (indirect seasonal factors), "cis" (indirect period-to-period changes).
main_title	Title for the plot. Default is character string 'Maximum Percent Difference Comparison Plot'.
sub_title	Subtitle for the plot. Optional entry.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "

do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
cut_color	Color use to show the slidings spans cut off for this type of series. Default is "red". For a listing of R color codes, see R Coder (2025).
this_legend_title	Character string; indicates title of legend. Default is 'Sliding Spans'.
this_legend_entry	Character array; entries for the legend. Default is c("ss1", "ss2").
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.
this_legend_text_size	integer scalar; Size of the legend title. Default is 10.
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 2.6.1, 2026-06-09

Value

A ggplot object that generates a plot of the maximum percent difference from a sliding spans analysis of seasonal factors or changes.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

- Findley DF, Monsell BC, Shulman HB, Pugh MG (1990). "Sliding Spans Diagnostics for Seasonal and Related Adjustments." *Journal of the American Statistical Association*, **85**, 345–355. doi:10.2307/2289770.
- R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].
- Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.
- U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```

shoes_sspan_seas <-
  seasonal::seas(blsplotGG::shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    slidingspans.save = c('sfs', 'chs'))

shoes_sspan_x11_seas <-
  seasonal::seas(blsplotGG::shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    x11.seasonalma = "s3x3",
    slidingspans.length = 76,
    slidingspans.numspans = 4,
    slidingspans.print = 'sfs',
    slidingspans.save = c('sfs', 'chs'))
p_shoes_maxpct_sf <-
  plot_sliding_spans_two(shoes_sspan_seas,
    shoes_sspan_x11_seas, this_series = "sfs",
    main_title = "Maximum Difference of the Seasonal Factors",
    sub_title = "US Retail Shoe Sales",
    cut_color = "purple",
    this_legend_entry = c("Seats", "X-11"))

```

plot_spectrum	<i>Generate spectrum plot of either the original, seasonally adjusted, irregular, or model residuals.</i>
---------------	---

Description

Generate plot of spectrum from X-13ARIMA-SEATS specified by the user.

Usage

```

plot_spectrum(
  seas_obj = NULL,
  this_spectrum = "sp0",
  xaxis_bls = TRUE,
  main_title = "AR Spectrum",
  sub_title = NULL,
  series_name = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  line_color = "darkblue",
  median_color = "blue",
  freq_color = c("steelblue", "forestgreen"),
  peak_color = c("violet", "brown")
)

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_spectrum	Character string; three character code for the X-13 spectrum to be generated. Allowed entries are "sp0" (modified original series), "sp1" (modified X-11 seasonally adjusted series), "sp2" (modified X-11 irregular), "s1s" (modified SEATS seasonally adjusted series), "s2s" (modified SEATS irregular), "is0" (modified composite series), "is1" (modified indirect seasonally adjusted series), "is2" (modified indirect irregular), spr (model residuals), or "ser" (extended residuals). Default: "sp0".
xaxis_bls	Logical scalar; indicates if x-axis of spectral plot will be frequency by month rather than the actual frequencies. Default sets x-axis to frequency by month.
main_title	Character string; main title of plot. Default is 'AR Spectrum'.
sub_title	Character scalar; Description of time series used in seas_obj. Used as the subtitle of the plot if specified.
series_name	Character scalar; name of the time series used in seas_obj. Used as the label of the Y-axis if specified.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character string. Colors used for spectrum in plot. Default is "darkblue". For a listing of R color codes, see R Coder (2025).
median_color	Character string. Colors used for medians of the spectrum. Default is "blue".
freq_color	Character vector of length 2. Colors used for seasonal and trading day frequencies, respectively. Defaults are c("steelblue", "forestgreen").
peak_color	Character vector of length 2. Colors used for visually significant peaks at seasonal and trading day frequencies, respectively. Visually significant peaks determined using the criteria of Soukup and Findley (1999). Defaults are c("violet", "brown").

Details

Version 3.2, 2026-06-09

Value

A ggplot object which generates a spectrum plot generated by X-13ARIMA-SEATS.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of*

Statistical Software, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Soukup RJ, Findley DF (1999). “On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment.” *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. <https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf>.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = "(0 1 1)(0 1 1)", x11="",
    spectrum.save = "sp2")
plot_air_spectrum <-
  plot_spectrum(air_seas, this_spectrum = "sp2",
    series_name = "AirPassengers",
    sub_title = "Airline Passengers",
    line_color = "steelblue", median_color = "blue",
    freq_color = c("darkblue", "darkgreen"),
    peak_color = c("red", "orange"))
```

plot_squared_gain	<i>Plot of the squared gains for filters generated by an X-13ARIMA-SEATS SEATS seasonal adjustment run.</i>
-------------------	---

Description

Generate squared gains plot of the concurrent and symmetric SEATS seasonal adjustment and trend filters from a SEATS adjustment from a seas object generated by the seasonal package. For more information on SEATS and SEATS input options in X-13ARIMA-SEATS, (see Section 7.14 of U. S. Census Bureau 2025).

Usage

```
plot_squared_gain(
  seas_obj = NULL,
  this_series = "sa",
  main_title = NULL,
  sub_title = NULL,
  this_y_label = "Squared Gain",
  this_x_label = "Cycles per Year",
  do_grid = FALSE,
  do_background = FALSE,
  line_color = NULL,
  this_palette = "Paired",
  this_legend_title = "Filter",
  legend_title_size = 12,
  legend_text_size = 10
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_series	Character string; series for which SEATS produces a squared gain plot, limited to the seasonally adjusted series (sa, the default), or the trend component (trend). For other entries, the function will print an error message and return a NULL.
main_title	Character string; main title of plot. A title will be generated if no title is specified.
sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot. Default is "Squared Gain".
this_x_label	Label for X axis. Default is "Cycles per Year".
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character vector of length two; colors used for the squared gain in the plot. For a listing of R color codes, see R Coder (2025). Default is NULL, which indicates that the palette specified in this_palette is used to generate colors for this plot.
this_palette	Color used for lines in plot. Default is the RColorBrewer palette "Paired" (Neuwirth 2022).
this_legend_title	Title for legend. Default is "Filter"
legend_title_size	integer scalar; Size of the legend title. Default is 12.
legend_text_size	integer scalar; Size of the legend title. Default is 10.

Details

Version 2.5.1, 2026-06-09

Value

A ggplot object which generates a plot of the squared gains for filters generated by the SEATS seasonal factors. If SEATS seasonal adjustment (with finite = yes) not produced, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsel@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed

17-02-2026].

Sax C, Eddebuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). “X-13ARIMA-SEATS Reference Manual, Version 1.1.” U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```
shoes_seats_seas <-
  seasonal::seas(shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    seats.finite = "yes",
    seats.save = c( 'gac', 'gaf', 'gtc', 'gtf' ) )
p_sa_squared_gain <- plot_squared_gain(shoes_seats_seas,
  sub_title = "US Shoe Sales",
  this_palette = "Set2")
p_trend_squared_gain <- plot_squared_gain(shoes_seats_seas,
  sub_title = "US Shoe Sales",
  this_series = "trend",
  this_palette = "Set2")
```

plot_table

Plot table from X-13ARIMA-SEATS seasonal adjustment run.

Description

Generate plot of user-specified table from the output of a seas object generated by the seasonal package.

Usage

```
plot_table(
  seas_obj = NULL,
  this_table = NULL,
  main_title = NULL,
  sub_title = NULL,
  this_y_label = NULL,
  y_limit = NULL,
  this_x_label = "Time",
  start_plot = NULL,
  do_grid = FALSE,
  do_background = FALSE,
  draw_recess = FALSE,
  recess_color = "lightgrey",
  recess_sub = TRUE,
  add_outlier = FALSE,
```

```

use_ratio = FALSE,
ratio_mean = 1,
this_line_type = "solid",
line_color = "grey",
outlier_color = c("red", "blue", "orangered", "green", "steelblue", "blue"),
outlier_line_type = c("dashed", "dotdash", "dashed", "twodash", "dotdash", "dotdash")
)

```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_table	Character string; X-13ARIMA-SEATS table name or abbreviation. Valid table names for this argument can be found in the documentation for the series function of the seasonal package (Sax 2018). If not a valid table name, the function will print an error message and return a NULL.
main_title	Character string; main title of plot. A title will be generated if no title is specified.
sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot, if specified.
y_limit	Numeric vector of length 2; Range of values for the y-axis. Default is range of the series specified.
this_x_label	Label for X axis. Default is "Time".
start_plot	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates. Default is no recession shading.
recess_color	Character string; color used for shading of recession region. Default is 'lightgrey'.
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label.
add_outlier	Logical scalar; indicates if lines for identified outliers are included in series plots. Default is not including lines for identified outliers.
use_ratio	Logical scalar; indicates if plots of seasonal factors, irregular, and residuals are done as ratio plots. Default has these plots as time series line plots.
ratio_mean	Assumed mean value for the ratio. Default is 1.0
this_line_type	Character string; indicates line type of each plot produced. Default is "solid".
line_color	Character string; color used for series in the plot. Default is 'grey'. For a listing of R color codes, see R Coder (2025).
outlier_color	Character array of length 6; color used for different outliers, with the order being 'ao', 'ls', 'tc', 'so', 'rp', 'tls'. Default is c("red", "blue", "orangered", "green", "steelblue", "blue").
outlier_line_type	Character array of length 6; Line type used for different outliers, with the order being 'ao', 'ls', 'tc', 'so', 'rp', 'tls'. Default is c('dashed', 'dotdash', 'dashed', 'twodash', 'dotdash', 'dotdash').

Details

Version 3.5.1, 2026-06-09

Value

A ggplot object that generates a plot of user-specified series from an X-13ARIMA-SEATS table. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS. R package version 1.7*. <https://CRAN.R-project.org/package=seasonal>.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11="",
    series.save = 'b1', transform.function = "log",
    x11.save = "e3")
air_d11_p <- blsplotGG::plot_table(air_seas, "d11",
  this_y_label = "AirPassengers",
  main_title = "X-11 Seasonal Adjustment of Airline Passengers",
  sub_title = "Seasonal Adjustment (D11)",
  do_grid = TRUE, draw_recess = TRUE,
  use_ratio = FALSE, add_outlier = TRUE, line_color = "darkblue")
air_d16_p <- blsplotGG::plot_table(air_seas, "d16",
  this_y_label = "AirPassengers",
  main_title = "X-11 Seasonal Adjustment of Airline Passengers",
  sub_title = "Combined Factors (D16)",
  do_grid = FALSE, draw_recess = TRUE,
  use_ratio = TRUE, add_outlier = TRUE, line_color = "steelblue")
air_e3_p <- blsplotGG::plot_table(air_seas, "e3",
  this_y_label = "AirPassengers",
  main_title = "X-11 Seasonal Adjustment (Extreme Adjusted) of Airline Passengers",
  sub_title = "Box-Jenkins Airline series",
  do_grid = FALSE, draw_recess = TRUE,
  use_ratio = TRUE, add_outlier = TRUE, line_color = "steelblue")
```

plot_time_shift	<i>Time shift plot for finite concurrent filters generated by an X-13ARIMA-SEATS SEATS seasonal adjustment run.</i>
-----------------	---

Description

Generate time shift plot of the finite concurrent SEATS seasonal adjustment and trend filters from a SEATS adjustment from a seas object generated by the seasonal package. For more information on SEATS and SEATS input options in X-13ARIMA-SEATS, (see Section 7.14 of U. S. Census Bureau 2025).

Usage

```
plot_time_shift(
  seas_obj = NULL,
  main_title = NULL,
  sub_title = NULL,
  this_y_label = "Time Shift",
  this_x_label = "Cycles per Year",
  do_grid = FALSE,
  do_background = FALSE,
  line_color = NULL,
  this_palette = "Paired",
  this_legend_title = "Filter",
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
main_title	Character string; main title of plot. A title will be generated if no title is specified.
sub_title	Character string; subtitle of plot. There is no default subtitle.
this_y_label	Character string; y-axis label for plot. Default is "Time Shift".
this_x_label	Label for X axis. Default is "Cycles per Year".
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
line_color	Character vector of length two; colors used for the squared gain in the plot. For a listing of R color codes, see R Coder (2025). Default is NULL, which indicates that the palette specified in this_palette is used to generate colors for this plot.
this_palette	Color used for lines in plot. Default is the RColorBrewer palette "Paired" (Neuwirth 2022).

this_legend_title
 Title for legend. Default is "Filter"
this_legend_title_size
 integer scalar; Size of the legend title. Default is 12.
this_legend_text_size
 integer scalar; Size of the legend title. Default is 10.
this_legend_position
 character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space
 integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 2.4.1, 2026-06-09

Value

A ggplot object which generates a plot of the squared gains for filters generated by the SEATS seasonal factors. If SEATS seasonal adjustment (with `finite = yes`) not produced, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

U. S. Census Bureau (2025). "X-13ARIMA-SEATS Reference Manual, Version 1.1." U.S. Census Bureau, U.S. Department of Commerce, <https://www2.census.gov/software/x-13arima-seats/x13as/windows/documentation/docx13ashtml.pdf>.

Examples

```

shoes_time_shift <-
  seasonal::seas(shoes2008,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log",
    forecast.maxlead = 36,
    check.print = c( 'pacf', 'pacfplot' ),
    seats.finite = "yes",
    seats.save = c( 'tac', 'ttc' ) )
p_time_shift <- plot_time_shift(shoes_time_shift,
  sub_title = "US Shoe Sales",
  this_palette = "Set2")

```

plot_two_sa	<i>Compare two seasonal adjustments</i>
-------------	---

Description

Generates a ggplot object with a time series plot that compares two seasonal adjustments of the same series, optionally including the original series.

Usage

```
plot_two_sa(
  this_sa_one = NULL,
  this_sa_two = NULL,
  this_ori = NULL,
  main_title = "Compare X-11 and SEATS",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = " ",
  this_sa_text = c("X-11", "SEATS"),
  do_grid = FALSE,
  do_background = FALSE,
  line_color = NULL,
  this_palette = "Dark2",
  this_legend_title = "Series",
  this_legend_title_size = 12,
  this_legend_text_size = 10,
  this_legend_position = "right",
  this_legend_space = 10
)
```

Arguments

this_sa_one	Time series of the X-11 seasonal adjustment. This is a required entry.
this_sa_two	Time series of the SEATS seasonal adjustment. This is a required entry.
this_ori	Time series of the original series. Optional entry.
main_title	Title for the plot. Default is character string 'Comparison of Seasonal Adjustments'.
sub_title	Subtitle for the plot. Optional entry.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "
this_sa_text	Character vector of length 2; labels for different seasonal adjustments. Note that if this_ori is set, the label for that line will be set as "Ori". Default is c('X-11', 'SEATS').
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;

line_color	Character scalar; color used for plot. User should specify colors in the order of original series, first SA, and second SA if this_ori is set, first SA and second SA otherwise. Default is the RColorBrewer palette "Dark2". For a listing of R color codes, see R Coder (2025).
this_palette	Character string; default RColorBrewer palette. Default is the RColorBrewer palette "Dark2" (Neuwirth 2022).
this_legend_title	Title for legend. Default is "Series"
this_legend_title_size	integer scalar; Size of the legend title. Default is 12.
this_legend_text_size	integer scalar; Size of the legend title. Default is 10.
this_legend_position	character String; The position of the legend, which can be one of top, bottom, left, right. Default is right.
this_legend_space	integer scalar; Space between legend and plot in points. Default is 10.

Details

Version 3.7, 2026-02-17

Value

A ggplot object that generates a plot comparing two seasonal adjustments, trend, or factors.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    x11.seasonalma = "s3x5",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_seats_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_x11_sa <- seasonal::final(ukgas_x11_seas)
ukgas_seats_sa <- seasonal::final(ukgas_seats_seas)
```

```
ukgas_seats_v_x11_p <-
  plot_two_sa(this_sa_one = ukgas_x11_sa,
             this_sa_two = ukgas_seats_sa,
             main_title = "UK Gas Seasonal Adjustments",
             sub_title = "X-11 - Blue, SEATS - Green",
             line_color = c("steelblue", "forestgreen"))
```

```
plot_two_sa_comparison
```

Compare two seasonal adjustments

Description

Generates a ggplot object with a time series plot that compares two seasonal adjustments of the same series, with an optional comparison panel between the two adjustments.

Usage

```
plot_two_sa_comparison(
  this_sa_one = NULL,
  this_sa_two = NULL,
  do_comparison_plot = TRUE,
  overall_title = "Compare X-11 and SEATS",
  sa_one_title = "First SA Plot",
  sa_two_title = "Second SA Plot",
  comparison_title = "Comparison Plot",
  title_size = 10,
  do_grid = FALSE,
  do_background = FALSE,
  this_scale = "fixed",
  line_color = "steelblue",
  ratio_color = "grey"
)
```

Arguments

<code>this_sa_one</code>	Time series of the first seasonal adjustment. This is a required entry.
<code>this_sa_two</code>	Time series of the SEATS seasonal adjustment. This is a required entry.
<code>do_comparison_plot</code>	Logical scalar. If TRUE, a ratio/difference plot of the two seasonal adjustments will be generated from <code>this_sa_one</code> and <code>this_sa_two</code> , if the series have the same periodicity. Default is TRUE.
<code>overall_title</code>	Title for the combined plot. Default is NULL.
<code>sa_one_title</code>	Title for the first plot. Default is character string 'First SA Plot'.
<code>sa_two_title</code>	Title for the plot. Default is character string 'Second SA Plot'.
<code>comparison_title</code>	Title for the comparison plot, if requested. Default is character string 'Comparison Plot'.
<code>title_size</code>	integer scalar; Size of the component titles. The overall title size will be <code>title_size + 2</code> . Default is 10.

do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
this_scale	Character scalar; set the scale for the y-axis of the facet plots. Default is "fixed". To set axis to be different for the different plots, set this_scale = "free_y".
line_color	Character string; color used for lines in plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).
ratio_color	Character string; Line color for the comparison ratio plot. Deault is "grey".

Details

Version 2.3, 2026-02-17

Value

A ggplot object that generates a plot comparing two seasonal adjustments, with an optional comparison panel.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    x11.seasonalma = "s3x5",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_seats_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_x11_sa <- seasonal::final(ukgas_x11_seas)
ukgas_seats_sa <- seasonal::final(ukgas_seats_seas)
ukgas_seats_v_x11_p <-
  plot_two_sa_comparison(
    this_sa_one = ukgas_x11_sa,
    this_sa_two = ukgas_seats_sa,
    overall_title = "UK Gas Production",
    sa_one_title = "X-11 Seasonal Adjustment",
    sa_two_title = "SEATS Seasonal Adjustment",
    comparison_title = "Ratio of Seasonal Adjustments (X11 / SEATS)")
```

plot_two_sa_facet	<i>Compare two seasonal adjustments in a facet plot</i>
-------------------	---

Description

Generates a ggplot object with a time series facet plot that compares two seasonal adjustments of the same series, optionally including the original series.

Usage

```
plot_two_sa_facet(
  this_sa_one = NULL,
  this_sa_two = NULL,
  this_ori = NULL,
  main_title = "Compare X-11 and SEATS",
  sub_title = NULL,
  this_x_label = "Time",
  this_y_label = " ",
  line_color = "steelblue",
  this_sa_text = c("sadj1", "sadj2"),
  do_grid = FALSE,
  do_background = FALSE,
  this_scale = "fixed"
)
```

Arguments

this_sa_one	Time series of the first seasonal adjustment. This is a required entry.
this_sa_two	Time series of the second seasonal adjustment. This is a required entry.
this_ori	Time series of the original series. Optional entry.
main_title	Title for the plot. Default is character string 'Comparison of Seasonal Adjustments'.
sub_title	Subtitle for the plot. Optional entry.
this_x_label	Label for X-axis. Default is "Time"
this_y_label	Label for Y-axis. Default is " "
line_color	Color used for lines in plot. Default is "steelblue". For a listing of R color codes, see R Coder (2025).
this_sa_text	Character vector of length 2; labels for different seasonal adjustments. Note that if this_ori is set, the label for that line will be set as "Ori". Default is c("sadj1", "sadj2").
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
do_background	Logical scalar; indicates grey background included in plot. Default is no grey background;
this_scale	Character scalar; set the scale for the y-axis of the facet plots. Default is "fixed". To set axis to be different for the different plots, set this_scale = "free_y".

Details

Version 7.4, 2026-02-17

Value

A ggplot object that generates a facet plot comparing two seasonal adjustments, trends, or factors.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

R Coder (2025). “Colors in R — r-charts.com.” <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
ukgas_x11_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    x11.seasonalma = "s3x5",
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_seats_seas <-
  seasonal::seas(UKgas, series.period = 4,
    arima.model = "(0 1 1)(0 1 1)",
    transform.function = "log", forecast.maxlead = 20,
    check.print = c( 'pacf', 'pacfplot' ))
ukgas_x11_sa <- seasonal::final(ukgas_x11_seas)
ukgas_seats_sa <- seasonal::final(ukgas_seats_seas)
ukgas_two_sa_facet_p <-
  plot_two_sa_facet(this_sa_one = ukgas_x11_sa,
    this_sa_two = ukgas_seats_sa,
    main_title = "UK Gas Seasonal Adjustments",
    line_color = "forestgreen",
    this_sa_text = c("X-11", "Seats"))
```

plot_year_over_year *Plot year over year plot*

Description

Generate year over year plot of a user-specified ts object.

Usage

```
plot_year_over_year(
  this_series = NULL,
  main_title = NULL,
  sub_title = NULL,
  this_y_label = NULL,
  y_limit = NULL,
  this_x_label = NULL,
  start_plot = NULL,
  do_grid = FALSE,
  do_background = FALSE,
```

```

    line_color = NULL,
    this_palette = "Paired",
    detrend_series = FALSE,
    detrend_lowess = FALSE,
    this_legend_title = "Years",
    this_legend_position = "right",
    this_legend_size = NULL,
    this_legend_order = NULL,
    this_legend_space = NULL
  )

```

Arguments

<code>this_series</code>	Numeric matrix; columns of time series object to be plotted.
<code>main_title</code>	Character string; main title of plot. The default title is the name of the series passed to this function.
<code>sub_title</code>	Character string; subtitle of plot. There is no default subtitle.
<code>this_y_label</code>	Character string; y-axis label for plot, if specified.
<code>y_limit</code>	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the series specified.
<code>this_x_label</code>	Label for X axis. Default is "Month" or "Quarter".
<code>start_plot</code>	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
<code>do_grid</code>	Logical scalar; indicates if plots will have grid lines. Default is no grid lines.
<code>do_background</code>	Logical scalar; indicates grey background included in plot. Default is no grey background.
<code>line_color</code>	Character scalar; color used for plot. User should specify one color for each column of the matrix specified. Default is the RColorBrewer palette "Paired". For a listing of R color codes, see R Coder (2025).
<code>this_palette</code>	Character string; RColorBrewer palette used for this plot.. Default is the RColorBrewer palette "Paired" (Neuwirth 2022).
<code>detrend_series</code>	Logical scalar; indicates if the series plotted is to be detrended. Default is the original series is plotted.
<code>detrend_lowess</code>	Logical scalar; indicates lowess is used to generate the trend used to detrend the series. Default setting is loess is not used to detrend the series; detrending doesn't happen unless <code>detrend_series = TRUE</code> .
<code>this_legend_title</code>	Character string; sets the title associated with the plot legend. Default title is "Years".
<code>this_legend_position</code>	Character string; indicates the position of the plot legend. Possible values are "right" (default), "left", "top", "bottom", and "none"
<code>this_legend_size</code>	Numeric scalar; indicates the size of several attributes of the plot legend, in centimeters. Default value is 1.
<code>this_legend_order</code>	Character vector; indicates the order of entries in the legend. Default order is determined by ggplot2.
<code>this_legend_space</code>	integer scalar; Space between legend and plot in points. Default is determined by ggplot2.

Details

Version 7.2, 2026-02-17

Value

Generate ggplot object generating a year to year plot of a time series object. If time series object not specified, print out error message and return NULL. Note that this plot will likely produce warning messages - to suppress them, use `suppressWarnings` when printing out the plot.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Examples

```
this_yyplot <-
  plot_year_over_year(AirPassengers, this_y_label = "Air",
    this_palette = "Dark2")
legend_order <-
  as.character(as.vector(matrix(1949:1960, ncol = 4, byrow = TRUE)))
this_yyplot_detrend <-
  plot_year_over_year(AirPassengers, this_y_label = "Air",
    this_palette = "Dark2",
    main_title = "YY Plot of Loess Detrended Air Passenger Series",
    detrend_series = TRUE, detrend_lowess = TRUE,
    this_legend_position = "bottom",
    this_legend_size = 0.25,
    this_legend_order = legend_order,
    this_legend_space = 0)
```

proc_outlier

Extract dates from outlier text

Description

Process name of outlier regressor to extract the dates associated with the outlier.

Usage

```
proc_outlier(this_outlier = NULL, this_freq = 12, add_type = TRUE)
```

Arguments

this_outlier	Character string; outlier regressor. This is a required entry.
this_freq	integer scalar; time series frequency. Default is 12.
add_type	logical scalar; determines if type of outlier is added to the output. Default is TRUE.

Details

Version 3.1, 2025-12-30

Value

List of either year and month/quarter of outlier, or year and month/quarter of start and end of outlier

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, x11="", slidingspans = "",
    transform.function = "log", arima.model = "(0 1 1)(0 1 1)",
    regression.aictest = 'td', forecast.maxlead=60,
    check.print = c( "pacf", "pacfplot" ))
this_auto_outlier <- get_auto_outlier_string(air_seas)
this_outlier      <- proc_outlier(this_auto_outlier)
```

set_day_and_month	<i>generate date object from days and year</i>
-------------------	--

Description

Generate a date object from the number of days from the beginning of the year and the target year. Uses the lubridate package Grolemund and Wickham (2011)).

Usage

```
set_day_and_month(day_number, target_year)
```

Arguments

day_number	Integer scalar, number of days from the start of the year. This is a required entry.
target_year	Integer scalar, year of the date object. This is a required entry.

Details

Version 1.1.1, 2026-06-18

Value

Date object determined by the input arguments.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Grolemund G, Wickham H (2011). “Dates and Times Made Easy with lubridate.” *Journal of Statistical Software*, **40**(3), 1–25. <https://www.jstatsoft.org/v40/i03/>.

Examples

```
this_start <- set_day_and_month(1, 2001)
this_end   <- set_day_and_month(350, 2025)
```

shoes2008	<i>Retail sales of shoes, 2008</i>
-----------	------------------------------------

Description

A time series object containing retail sales of shoes from the US Census Bureau.

Usage

```
data(shoes2008)
```

Format

Retail sales of shoes ending in April of 2008.

update_fts	<i>Update final t-statistics from outlier identification</i>
------------	--

Description

Update final t-statistics from automatic outlier identification with t-statistics from identified outliers in final regARIMA model.

Usage

```
update_fts(seas_obj = NULL)
```

Arguments

`seas_obj` seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.

Details

Version 1.3, 2026-06-09

Value

Updated fts matrix.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <- seasonal::seas(AirPassengers,
  arima.model = "(0 1 1)(0 1 1)",
  forecast.maxlead = 60, forecast.save = "fct",
  outlier.save = "fts")
air_fts <- seasonal::series(air_seas, "fts")
air_fts_updated <- update_fts(air_seas)
```

update_vector	<i>Update vector.</i>
---------------	-----------------------

Description

Fill unspecified elements of a vector with the first element of the input series.

Usage

```
update_vector(this_series = NULL, this_num = NULL)
```

Arguments

- this_series Original time series. This is a required entry.
- this_num Length of updated series. Must be more than the length of this_series. This is a required entry.

Details

Version 2.3, 2023-05-25

Value

An updated vector of length this_num augmented with the first value of the input series.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this_vector <- c(1,2)
updated_vector <- update_vector(this_vector, 4)
```

visual_sig_peaks	<i>Flag visual significant peaks in spectra</i>
------------------	---

Description

Determine positions of visual significant peaks in spectra.

Usage

```
visual_sig_peaks(seas_obj = NULL, spec_type = "sa", spec_freq_code = "seas")
```

Arguments

seas_obj	seas object for a single series generated by the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
spec_type	Character string; type of spectrum. Possible values are 'ori', 'irr', 'rsd', 'sa', 'comp', 'indsa', 'indirr', 'extrsd'. Default is 'sa'.
spec_freq_code	Character string; type of frequency being tested. Possible values are 'seas' or 'td'. Default is 'seas'.

Details

Version 4.1.1, 2026-06-09

Value

If visually significant peaks found using the criteria of Soukup and Findley (1999), a numeric vector of the position of the peak frequencies. If no peaks found, this function returns 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Soukup RJ, Findley DF (1999). “On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment.” *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. <https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf>.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)', x11='')
vp_ori_seas <- visual_sig_peaks(air_seas, spec_type = 'ori')
vp_ori_td <-
  visual_sig_peaks(air_seas, spec_type = 'sa',
    spec_freq_code = 'td')
```

what_spectrum	<i>What spectrum is plotted</i>
---------------	---------------------------------

Description

Return which spectrum plot is generated based on the three character code used by the `plot_spectrum` function.

Usage

```
what_spectrum(this_spectrum = "sp0", use_title_case = FALSE)
```

Arguments

- `this_spectrum` Character string; three character code for the X-13 spectrum to be generated. Allowed entries are "sp0" (modified original series), "sp1" (modified X-11 seasonally adjusted series), "sp2" (modified X-11 irregular), "s1s" (modified SEATS seasonally adjusted series), "s2s" (modified SEATS irregular), "is0" (modified composite series), "is1" (modified indirect seasonally adjusted series), "is2" (modified indirect irregular), spr (model residuals), or "ser" (extended residuals). Default: "sp0".
- `use_title_case` Logical scalar; convert string to title case. Default is FALSE.

Details

Version 1.5, 2024-05-06

Value

Text for spectrum associated with code used in `plot_spectrum` function. If improper value set for `this_spectrum`, function will return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
sub_title <- what_spectrum("sp2", use_title_case = TRUE)
```

xt_data_list*US Building Permits, One Family Units*

Description

A list object of US One family Building Permits from the US Census Bureau for four regions expressed as time series objects that end in October, 2006.

Usage

```
data(xt_data_list)
```

Format

A list object with 4 time series elements:

mw1u Midwest one family building permits

ne1u Northeast one family building permits

so1u South one family building permits

we1u West one family building permits

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